Robotic Manipulator Device Control Unit
ROMACU 2
Firmware Command Set

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1 Document History

Version	Date	Author	Description of document changes	Corresponding Firmware
0.1	16.09.2003	HARO	Initial draft of this document based on the former generation ROMACU Doc. No. 391 973 V1.13.	ROMACU V2.00
0.2	19.09.2003	HARO	Corrections according Reviewbericht ROMA2_FCS_Review20030917.doc.	ROMACU V2.00
1.0	13.10.2003	HARO	Final release version	ROMACU V2.00
1.10	2005-10-20	ROST	Change document name from FCS_ROMACU2V1.0.doc into FCS_ROMACU2.doc <RDR> change unit of measurement Update error codes and trace ability Insert <REB>, <BRM>, <BSM>, <TRB>, <ARC>, <ABR> and <RIB> commands Insert Axis description and Overview BCS	ROMACU V2.10
1.20	2007-05-08	Roman Steiner	Changes on <RSM, <SSM>, <RAA> and <SAA> commands. New commands: <ACW>, <Txx>, <RTX>, <SPC>, <RPC>, <STW>, <RTW>, <SPT>, <RPT>, <AAC>, <SMD>, <RMD> and <MMD>	ROMACU V2.20

2 Firmware Update History

FW-Version	Description of functional changes
ROMACU-V2.00	- Main difference to the former generation ROMACU-V1.13 is the exchange of the old DC-Servo controller by the new DC-Servo 2 module. This new Version V2.00 is working only with the new DC-Servo 2 modules. - Increased acceleration in Z direction.
ROMACU-V2.10	- Enhance error information in <REE> (Bug tracker #132) - Insert interruptible initialization routines (Bug tracker #138) - Insert Barcode-Scanner functionalities (Bug tracker #254) - Adaptation of X-axis acceleration (Bug tracker #264)
ROMACU-V2.20	- Correct autorange for small scale factors (Bug tracker #283) - Release AMD, Macronix and ST flash devices (Bug tracker #299) - Enable round curves(Bug tracker #316) - Enhance error information in <REE> (Bug tracker #275) - Enable multiple download for subdevices (Bug tracker #317)

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3 Introduction

3.1 Purpose

This document contains detailed information about all firmware commands implemented in the firmware of the ROMACU 2. It also assures the upward and downward traceability between this document and the firmware requirements document.

3.2 Scope

This document is intended for software engineers in order to write service and setup tools or application software. It is not part of the end-user documentation.

3.3 Abbreviations

The document uses the following abbreviations:

Abbreviation	***Definition***
CAN	Controller Area Network.
DCU	Device Control Unit.
FLASH EPROM	Electrical Erasable Programmable Read Only Memory.
FCS	Firmware Command Set
FRS	Firmware Requirements Specifications
FW	Firmware.
HCAN	Highest level CAN bus (CAN bus level TeCU -> DCU).
HDI	HITACHI Debugging Interface.
HEW	HITACHI Embedded Workshop.
LCAN	Local level CAN (CAN bus level DCU -> DC-Servo 2).
RAM	Random Access Memory.
RSP	Robotic Sample Processor.
RTX	Real Time eXecutive.
TeCU	Master Control Unit of an Instrument.
USB	Universal Serial Bus.
BCS	Barcode Scanner

3.4 Document Conventions

The document uses the following convention:

- Parameters marked with an asterisk denote parameters that may be omitted. Such omitted parameters retain the current setting or position.
- Omitted parameters not marked with an asterisk, assume the value 0 (zero).

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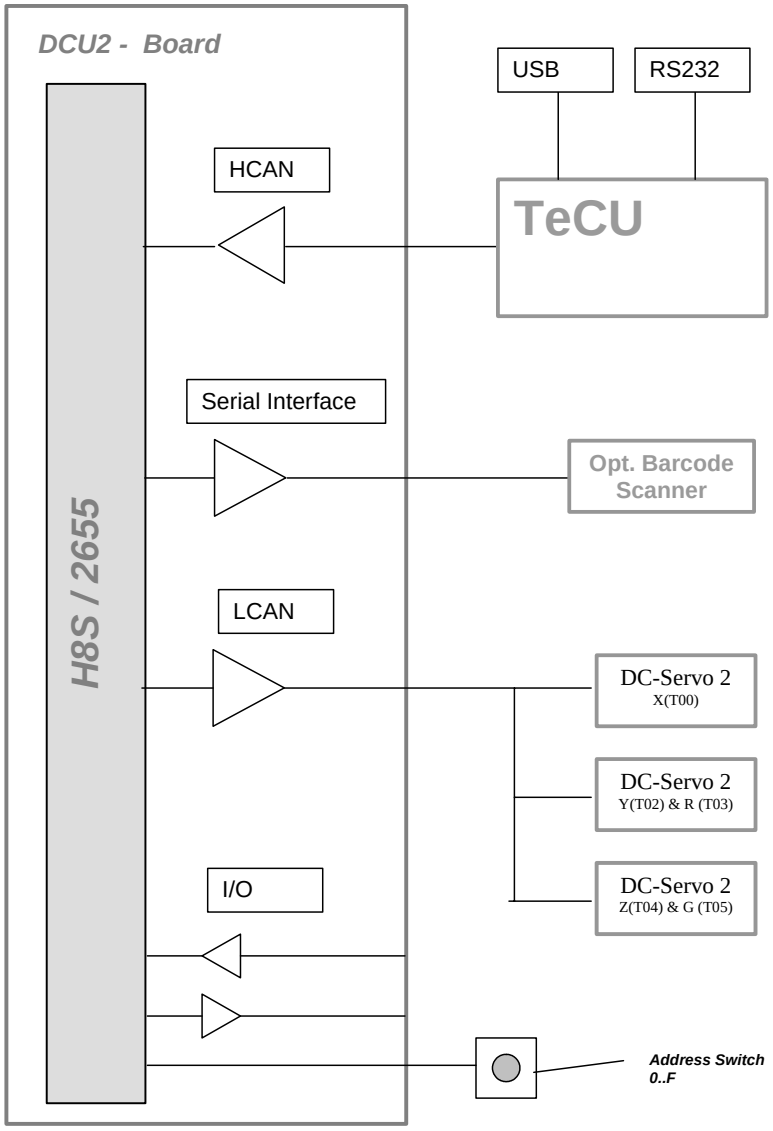
3.5 Device Overview

The DCU2 is using two identical general-purpose standalone Phillips CAN controllers called SJA1000. One of the interface controllers is used for the upper layer communication (HCAN) to the TeCU. The other controller is used for the LCAN interface to the lower level DC-Servo 2 drives.
In the ROMA FW, one RS interface is used to control the optional BCS.
Supplementary the DCU2 has two further RS interfaces for future expansion and several input and output lines.



3.6 Operation & Setup

Preliminary actions have to be performed in order to operate the module.

A. Installation of bootware
B. Assure correct CAN-Bus termination (60 Ohm)
C. Set address switch to the correct address
D. Installation of firmware
E. Hardware configuration
F. Calibration

Boot loading

The boot itself cannot be loaded via the CAN interface. It has to be programmed on a dedicated flash programmer *before* the assembly.

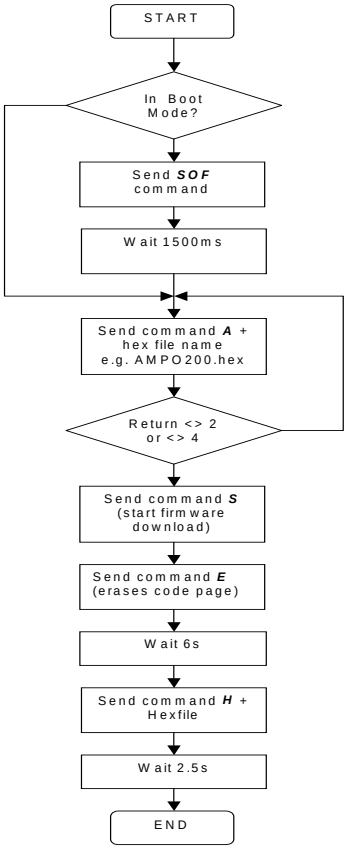
Firmware Download / Boot commands

The firmware download is initiated with an ***SOF*** command if there is already a valid firmware running. The program now jumps into the boot mode.

The boot program supports the following commands:

- ***'A'*** followed by the hex file name checks if the first three characters of the hex filename do correspond with the three characters <'xxx'>. This protects the module from loading a firmware that belongs to another device.
- ***'S'*** initiates the firmware download process. It applies the program voltage for the FLASH memory and copies the download procedure into the RAM memory range.
- The command ***'E'*** erases Flash application page.
- ***'H'*** followed by a bytestream writes directly into the FLASH memory. A 'EOF' string indicates the end of the Hexfile. A redundancy checksum is calculated and compared with a desired value. The program jumps into the firmware page and executes the downloaded code. In case of a deviation between checksums the program remains in the boot mode.
- The command ***'L'*** reads the type of equipment.
- ***'N'*** reads the number of nodes
- ***'X'*** exits boot mode.

firmware download sequence

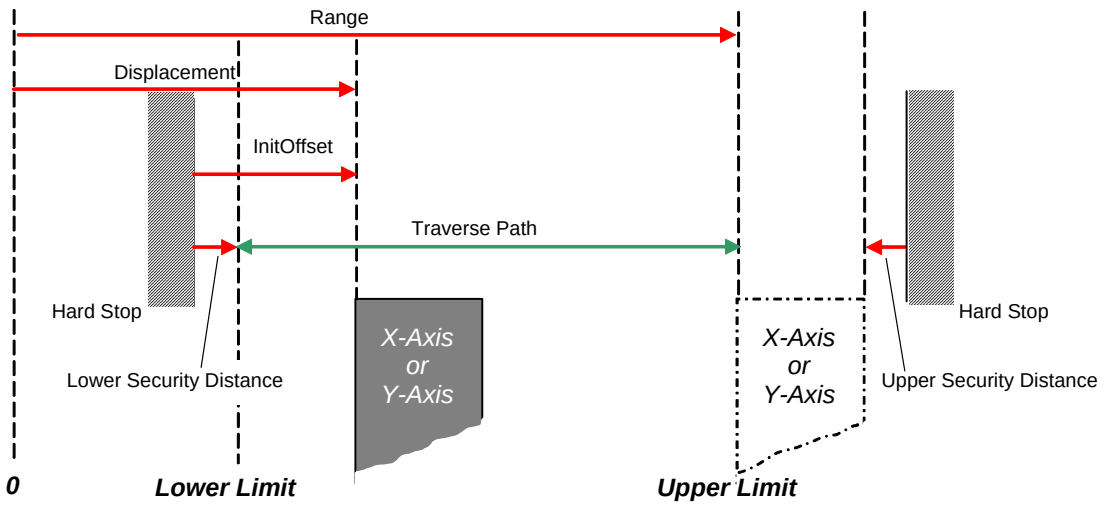


3.7 Axes Conventions

Life cycle is guarantied with default speed and acceleration settings.

Overview X- and Y- Axis

The X- and the Y-Axis work with an identical coordination system.



Upper Limit = Range
Lower Limit = Displacement – InitOffset + Lower Security Distance
Traverse Path = Upper Limit – Lower Limit

X-axis	***Parameter***	***Values***	***Units***
	Default Displacement	*0*	$^{1}/_{10}$mm
	Default InitOffset	*10*	$^{1}/_{10}$mm
	Default Range	*1'000*	$^{1}/_{10}$mm
	Upper Security Distance	*10*	$^{1}/_{10}$mm
	Lower Security Distance	*10*	$^{1}/_{10}$mm

Y-axis	***Parameter***	***Values***	***Units***
	Default Displacement	*-620*	$^{1}/_{10}$mm
	Default InitOffset	*10*	$^{1}/_{10}$mm
	Default Range	*1'000*	$^{1}/_{10}$mm
	Upper Security Distance	*10*	$^{1}/_{10}$mm
	Lower Security Distance	*10*	$^{1}/_{10}$mm

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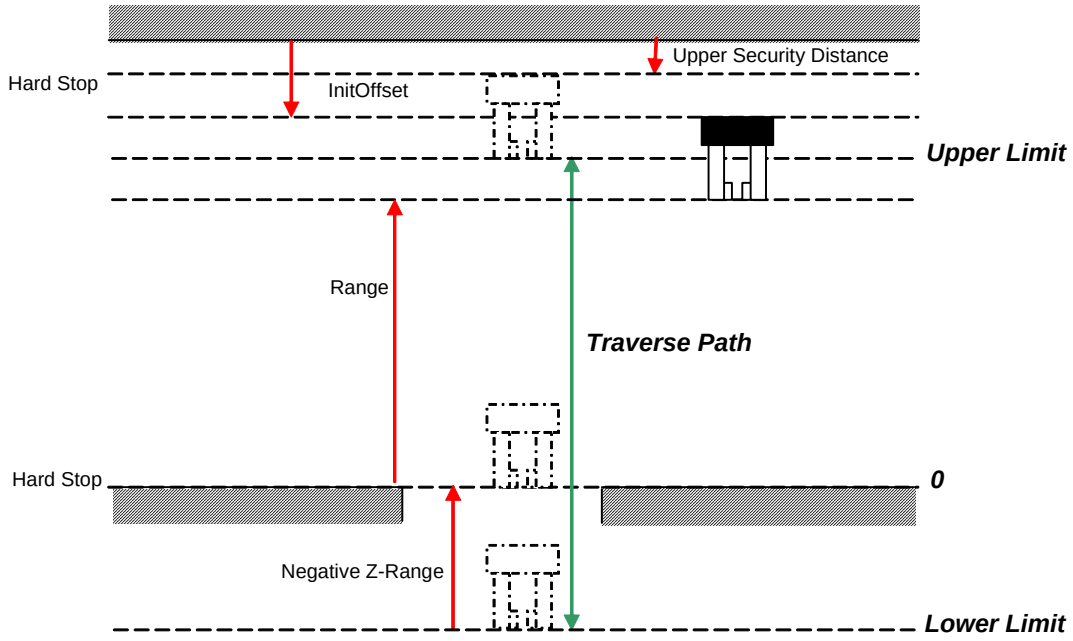
Overview Z-Axis

The Z-Axis initializes to the upper Hard Stop position and will be placed at the InitOffset position from there. The Zero-position is set on the worktable.

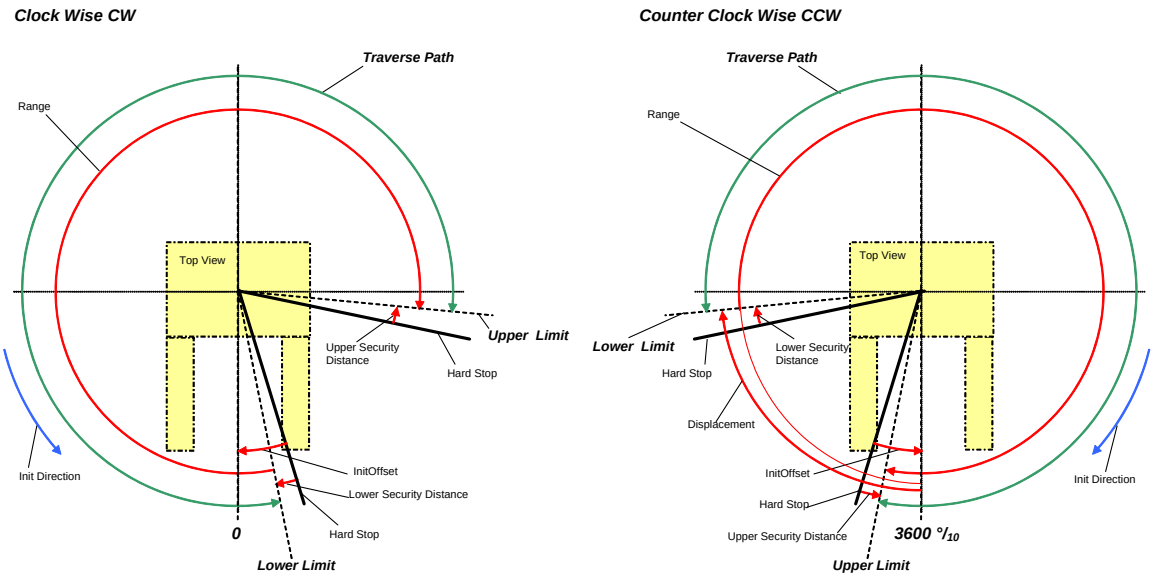


Upper Limit = Range + InitOffset – Upper Security Distance
Lower Limit = Negative Z-Range
Traverse Path = Range - Negative Z-Range + InitOffset

Z-axis	***Parameter***	***Value***	***Unit***
	Default Negative Z-Range	*0*	$^{1}/_{10}$mm
	Default InitOffset	*10*	$^{1}/_{10}$mm
	Default Range	*1000*	$^{1}/_{10}$mm
	Upper Security Distance	*10*	$^{1}/_{10}$mm

Overview R-Axis

The R-Axis is able to rotate about 270°. It can be driven in Clockwise Mode (0-270°) or in Counter Clock Wise Mode (90-360°).



Clock Wise (CW):

Upper Limit = Range
Lower Limit = Displacement – InitOffset + Lower Security Distance
Traverse Path = Upper Limit – Lower Limit

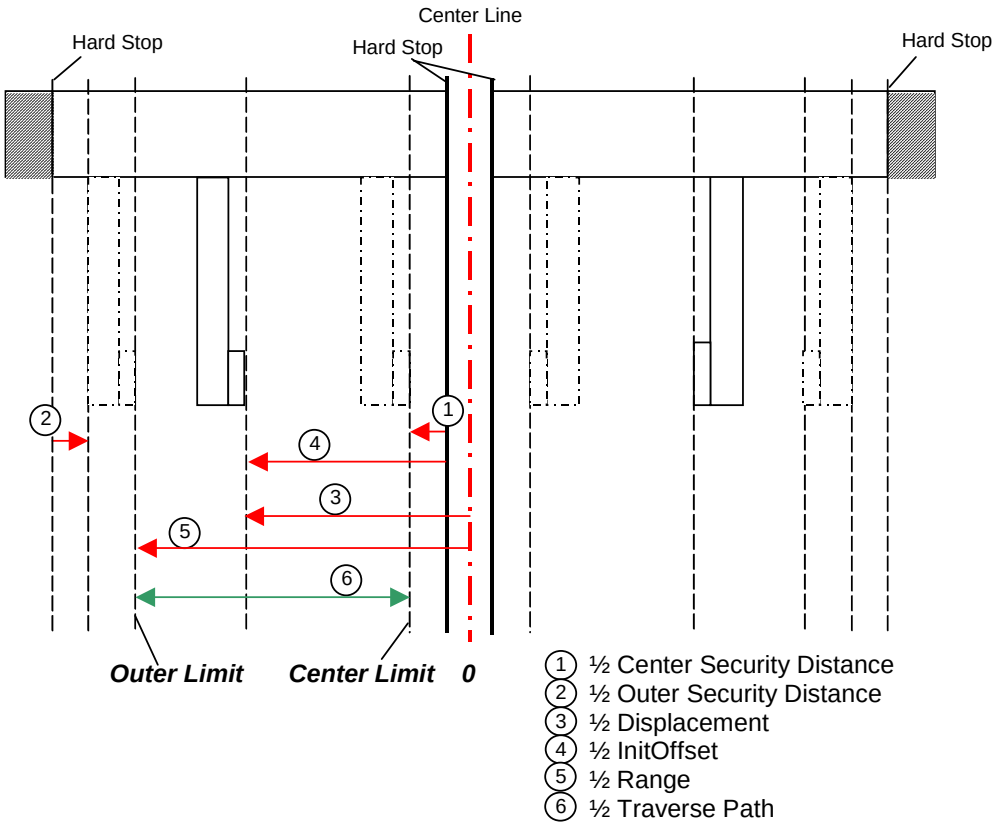
Counter Clock Wise (CCW):

Upper Limit = Range
Lower Limit = Displacement
Traverse Path = Upper Limit – Lower Limit

R-axis	***Parameter***	***Value***		***Unit***
		CW	***CCW***	
	Default Displacement	*0*	*900*	$^{1}/_{10}$°
	Default InitOffset	*50*	*50*	$^{1}/_{10}$°
	Default Range	*1'000*	*1'000*	$^{1}/_{10}$°
	Security Distance	*10*	*10*	$^{1}/_{10}$°

Overview G-Axis

Both fingers of the gripper work simultaneously to the opposite direction.



Outer Limit = Range
Center Limit = Displacement - InitOffset + Center Security Distance
Traverse Path = Range - Center Limit

G-axis	***Parameter***	***Value***	***Unit***
	Default Displacement	*560*	$^{1}/_{10}$mm
	Default InitOffset	*20*	$^{1}/_{10}$mm
	Default Range	*1560*	$^{1}/_{10}$mm
	Center Security Distance	*10*	$^{1}/_{10}$mm
	Outer Security Distance	*10*	$^{1}/_{10}$mm

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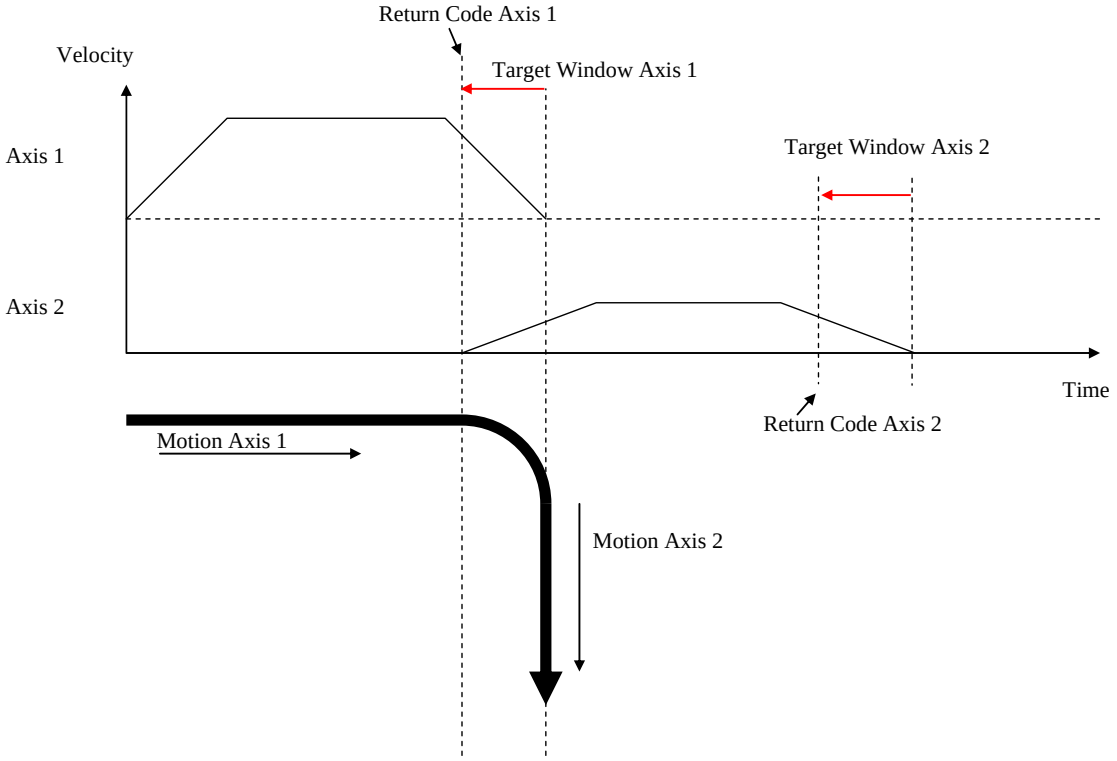
3.8 Round Corners with Target Windows

Round Corners in Roma can be achieved by starting the second motion prematurely.
The Target Window defines the position of the first axis, when the second axis has to be started.

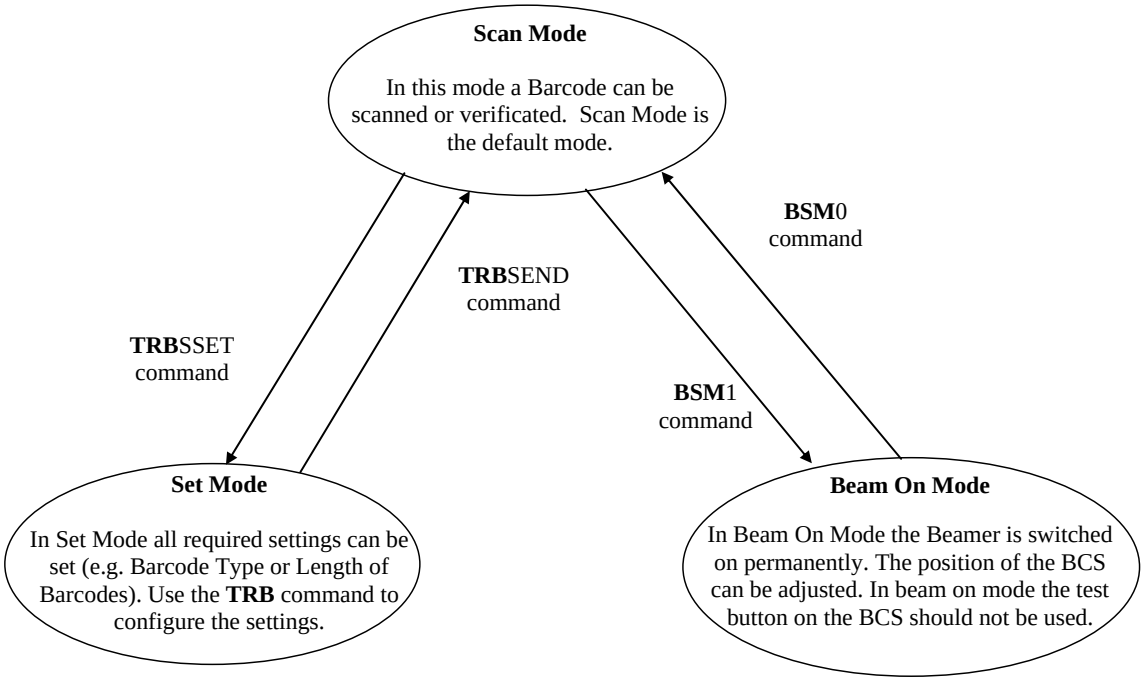
The agminated shape (radius and symmetry) of the entire motion depends of the predefined target window, speed and acceleration settings.



3.9 Barcode Scanner

The Barcode Scanner is annexed to the FW by a serial communication interface. This FW provides a communication pipeline straight to the BCS. It is possible to work with the BCS completely using this pipeline, but there are also commands implemented to handle more abstract functionalities like scanning and polling a barcode or keep turned on the beamer. With these special commands the required timeouts and buffer handling are automatically managed.

The Barcode Scanner works with three different operation modes:



For further details of the BCS specific commands for **TRB** command see the correlatively command set of the barcode scanner: BL-600 user's manual by Keyence. The **TRB** command can be found later in this document.

4 Commands

4.1 Report Commands (Base Functionality)

RFV *Report Firmware Version:*

RFV Selector			
Reports several systems depending information about the actual loaded firmware or hardware. This information can be used to identify the characteristics of the selected device.			
Parameter	***Validity***	***Default***	***Description***
Selector	0.. 5, 7, 9, 10, 15, 16	0	Report selected system information.
Response	***if***		***Range***
Selector	0		Report actual downloaded application firmware version. Structure: 'ROMACU-V2.xx-MM/YYYY'
	1		Report programmed bootstrap version.
	2		Report instrument serial number.
	3		Report actual hardware version of installed DeviceCU board.
	4		Report CAN bus node selection port.
	5		Report TECAN copyright.
	7		Report the number of nodes used by this device.
	9		Report actual DCU library version.
	10		Report expected HEX file name for firmware download. Structure: 'DCURO'
	15		Report application firmware as integer, V1.00 = 100.
	16		Report expected DC-Servo2 firmware version.
Error	(3) Invalid operand.		

RDA *Report Device Allocation Array of System Devices:*

RDA Layer, Type			
Reports the number of devices connected to the system in binary or decimal format. The system is divided in to eight layers, where each of them containing sixteen device nodes.			
Parameter	***Validity***	***Default***	***Description***
Layer	0..7	0	Report system devices of selected layer.
Type	0	0	Report value in binary format.
	1		Report value in decimal format.
Response	***if***		***Range***
Type	0		0..65535
	1		0..16
Error	(3) Invalid operand.		

RDF *Report Diagnostic Functions:*

RDF Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..3	0	Reports selected diagnose function.
Response	***if***		***Range***
Selector	0		Report the number of power ups.
	1		Report the overall operating time in minutes.
	2		Report the number of firmware downloads in Flash-EPROM.

	3		Report the number of parameter page erasing in Flash-EPROM.
Error	(3) Invalid operand.		

RGD *Report Global Data:*

RGD Selector			
This command reads the selected global data from the volatile memory. A maximum of 100 variables (type of integer) can be stored with the **SGD** command. CAUTION! This command is for internal use only (TECAN software). Overwriting of such variables can cause malfunction of the instrument and software.			
Parameter	***Validity***	***Default***	***Description***
Selector	1..100		Report the selected global data value.
Response	***if***		***Range***
Selector	1..100		-32768..32767
Error	(3) Invalid operand.		

RED *Read FLASH-EPROM Data:*

RED Selector			
This command reads all the selected data from the Flash-EPROM mirrored in the volatile memory. To get an actual copy of the Flash-EPROM contents, an **ARE** command must be sent before reporting the data with the **RED** command. The size of this data array has not a fixed length. It depends on the actual loaded firmware version. The length of the data array can be reported with the selector 0. The reported data is always type of integer. CAUTION! This command is for service purposes only (board exchange). Overwriting of such variables can cause malfunction of the instrument and software.			
Parameter	***Validity***	***Default***	***Description***
Selector	0	0	Report overall Flash-EPROM size in number of integers.
	1..**RED0**		Report value of type integer.
Response	***if***		***Range***
Selector	1..**RED0**		-32768..32767
Error	(3) Invalid operand.		

RSU *Report Stack Usage:*

RSU TaskID, Selector			
This command reports the selected task name and the corresponding depth of the specified task. CAUTION! This command is for service purposes only.			
Parameter	***Validity***	***Default***	***Description***
TaskID	1..50		Select task number.
Selector	0..1	0	Select information type.
Response	***if***		***Range***
TaskID	1		Report the “MCANTx” task.
	2		Report the “MCANRx” task.
	3		Report the “MCANRep” task.
	4		Report the “SCANTx” task.
	5		Report the “SCANRx” task.
	6		Report the “MBoot” task.
	7		Report the “SBoot” task.
	8		Report the “SuperV” task.
	9		Report the “Intrpt” task.
	10		Report the “XPatch” task.
	11		Report the “Device” task.
	12		Report the “Device” task.
Selector	0		Reports the task name as string of maximal eight characters.
	1		Report the depth of the selected task.
Error	(3) Invalid operand.		

RTX *Report Real Time Executive Parameter:*

RTX Selector			
This command reports the real time executive parameters of the operating system. CAUTION! This command is for service purposes only.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..1	0	Select information type.
Response	***if***		***Range***
Selector	0		Report actual used memory blocks.
	1		Report the peak number of used memory blocks
Error	(3) Invalid operand.		

4.2 Report Commands (ROMA Specific Functionality)

RHW Report ROMA Hardware Version:

RHW			
This command reports the actual installed ROMA hardware version set by the **SHW** command. The following combinations are available.			
Parameter	***Validity***	***Default***	***Description***
Response	***If***		
	0		Undefined ROMA type, no PIA is possible.
	1		ROMA1 is not supported by this FW but in old GENESIS CU.
	2		Eccentric gripper type right side orientation. (CW)
	3		Eccentric gripper type left side orientation. (CCW)
	4		Centric gripper type right side orientation (Centrifuge Loader).(CW)
	5		Centric gripper type left side orientation (Centrifuge Loader). (CCW)
Error	None.		

ROD Report Gripper Outrigger Distance:

ROD			
This command reports the gripper outrigger distance either individually set by the **SOD** command or predefined set by the **SHW** command.			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
			0..32767 in 1/10mm.
Error	None.		

RAA Report Vector Coordinate Position:

RAA Vector			
The vector coordinate positions defined and set by the **SAA** command can be reported with this command. Answer format: X, Y, Z, R, G, slow/fast speed, ParamClass, WindowClass			
Parameter	***Validity***	***Default***	***Description***
Vector	1..100		Selects the Vector to be reported.
Response	***if***		
			Reports the XYZRG destinations in in ^{10}/mm resp. ^{10}/dg, the speed selection, the selected parameter class and the selected window class for the selected Vector.
Error	(3) Invalid operand.		

RPC Report Drive Parameter Classes for AAC:

RPC Class, ParamNo			
This command reports the parameters stored in the classes for AAC moves set by the **SPC** command. Answer format: Param X, Param Y, Param Z, Param R, Param G			
Parameter	***Validity***	***Default***	***Description***
Class	1..100		Selects the class to be reported.
ParamNo	0..3		Selects the parameter to be reported in the selected class.
Response	***If***		
ParamNo	0		Reports the Accelerations of ramp up of all axes stored in the selected class in $^{10}/mm_{/s^2}$ resp. $^{10}/dg_{/s^2}$.
	1		Reports the Accelerations of ramp down of all axes stored in the selected class in $^{10}/mm_{/s^2}$ resp. $^{10}/dg_{/s^2}$

	2		Reports the Slow Speeds of all axes stored in the selected class in ^{10}/mm resp. ^{10}/dg.
	3		Reports the Fast Speeds of all axes stored in the selected class in ^{10}/mm resp. ^{10}/dg.
Error	(3) Invalid operand.		

RTW *Report Target Window Classes for AAC:*

RTW Class			
This command reports the target windows stored in the classes for AAC moves set by the **STW** command. Answer format: Window X, Window Y, Window Z, Window R, Window G			
Parameter	***Validity***	***Default***	***Description***
Class	1..100		Selects the class to be reported.
Response	***If***		
			Reports the Target Windowa of all axes stored in the selected class in ^{10}/mm resp. ^{10}/dg.
Error	(3) Invalid operand.		

RPT *Report Permanent Target Window:*

RPT			
This command reports the permanent set Target Window of all axes. Answer Format: Winx, WinY, WinZ, WinR, WinG			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
			Report the permanent set Target Window
Error	(3) Invalid operand.		

RAE *Report Actual Vector Number of AAA or AAC Movement:*

RAE			
This command reports the actual driven vector number of an **AAA** or **AAC** movement. This can be done while running an **AAA** or **AAC** sequence by sending this command via the alternative access channel (second access node). Also in case of a movement error, the last vector number can be reported, when the ROMA has stopped execution.			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
			1..100 while execution or 0 if successful termination of the **AAA** or **AAC** command.
Error	None.		

REE *Report Extended Error Code:*

REE Selector			
This command gets the axis specific error codes. Possible are all the errors generated by the ROMA device. To visualize the error codes as ASCII characters, an offset of 40 hexadecimal is added to the error value. Note: When moving with target windows, erroneous moves will be detected after a time out of 1 second.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..1	0	
Response	***if***		
Selector	0		X, Y, Z, R, G error code i.e. @@@@@ if all axis are ok.
	1		'XYZRG' as string.

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Error	(3) Invalid operand.

RPX *Report Current Parameter for the X-Axis:*

RPX Selector			
This command reports the current parameter settings of the X-axis. Only one axis parameter can be reported at a time.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..11	0	Report selected X-axis parameter
Response	***if***		
Selector	0		Report current position in 1/10 mm.
	1		Report acceleration in 1/10 mm/s^2.
	2		Report fast speed in 1/10 mm/s.
	3		Report initialization speed in 1/10 mm.
	4		Report initialization offset in 1/10 mm.
	5		Report axis range in 1/10 mm.
	6		Report initialization deviation in encoder increments.
	7		Report displacement offset in 1/10 mm.
	8		Report scale adjust factor.
	9		Report slow speed in 1/10 mm/s.
	10		Report scale factor.
	11		Report target position in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem.		

RPY *Report Current Parameter for the Y-Axis:*

RPY Selector			
This command reports the current parameter settings of the Y-axis. Only one axis parameter can be reported at a time.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..11	0	Report selected Y-axis parameter
Response	***if***		
Selector	0		Report current position in 1/10 mm.
	1		Report acceleration in 1/10 mm/s^2.
	2		Report fast speed in 1/10 mm/s.
	3		Report initialization speed in 1/10 mm.
	4		Report initialization offset in 1/10 mm.
	5		Report axis range in 1/10 mm.
	6		Report initialization deviation in encoder increments.
	7		Report displacement offset in 1/10 mm.
	8		Report scale adjust factor.
	9		Report slow speed in 1/10 mm/s.
	10		Report scale factor.
	11		Report target position in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem.		

RPZ *Report Current Parameter for the Z-Axis:*

RPZ Selector			
This command reports the current parameter settings of the Z-axis. Only one axis parameter can be reported at a time.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..11	0	Report selected Z-axis parameter
Response	***if***		
Selector	0		Report current position in 1/10 mm.
	1		Report acceleration in 1/10 mm/s^2.
	2		Report fast speed in 1/10 mm/s.

	3		Report initialization speed in 1/10 mm.
	4		Report initialization offset in 1/10 mm.
	5		Report axis range in 1/10 mm.
	6		Report initialization deviation in encoder increments.
	7		Report displacement offset in 1/10 mm.
	8		Report scale adjust factor.
	9		Report slow speed in 1/10 mm/s.
	10		Report scale factor.
	11		Report target position in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem.		

RPR *Report Current Parameter for the R-Axis:*

RPR Selector			
This command reports the current parameter settings of the R-axis. Only one axis parameter can be reported at a time.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..11	0	Report selected R-axis parameter
Response	***if***		
Selector	0		Report current position in 1/10 dg.
	1		Report acceleration in 1/10 dg/s^2.
	2		Report fast speed in 1/10 dg/s.
	3		Report initialization speed in 1/10 dg.
	4		Report initialization offset in 1/10 dg.
	5		Report axis range in 1/10 dg.
	6		Report initialization deviation in encoder increments.
	7		Report displacement offset in 1/10 dg. (Always 0)
	8		Report scale adjust factor.
	9		Report slow speed in 1/10 dg/s.
	10		Report scale factor.
	11		Report target position in 1/10 dg/s.
Error	(3) Invalid operand, (4) CAN ack problem.		

RPG *Report Current Parameter for the G-Axis:*

RPG Selector			
This command reports the current parameter settings of the G-axis. Only one axis parameter can be reported at a time.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..11	0	Report selected G-axis parameter
Response	***if***		
Selector	0		Report current position in 1/10 mm.
	1		Report acceleration in 1/10 mm/s^2.
	2		Report fast speed in 1/10 mm/s.
	3		Report initialization speed in 1/10 mm.
	4		Report initialization offset in 1/10 mm.
	5		Report axis range in 1/10 mm.
	6		Report initialization deviation in encoder increments.
	7		Report displacement offset in 1/10 mm.
	8		Report scale adjust factor.
	9		Report slow speed in 1/10 mm/s.
	10		Report scale factor.
	11		Report target position in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem.		

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RGG *Report Gripper Parameters:*

RGG			
This command reports the gripper speed, PWM (pulse width modulation) limit and the maximal allowed current, set by the **SGG** command. These parameter settings are in use while gripping a plate with the **AGR** command.			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
			Speed, PWMLimit, CurLimit.
Error	None.		

RSM *Report X-Axis Smooth Move Mode:*

RSM Selector			
This command sets the X-axis smooth move and the YZRG-Axis Harmonization mode set by **SSM**.			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
Selector	0		0 = smooth move mode, 1 = normal speed parameter used.
	1		0 = YZRG speeds harmonized, 1 = normal set speed parameter used
Error	None.		

RBK *Report Z-Axis Break Status:*

RBK			
This command returns the actual status of the Z-axis break.			
Parameter	***Validity***	***Default***	***Description***
Response	***if***		
			0 = break active (power off), 1 = break released (power on).
Error	(4) CAN ack problem.		

RDX *Report Diagnostic Functions for the X-Axis:*

RDX Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2, 5	0	Reports selected diagnose function.
Response	***if***		
Selector	0		Report the number of move counter.
	1		Report the distance counter in meter.
	2		Report the no load (crash) counter.
	5		Report the clearable move counter.
Error	(3) Invalid operand.		

RDY *Report Diagnostic Functions for the Y-Axis:*

RDY Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2, 5	0	Reports selected diagnose function.
Response	***if***		
Selector	0		Report the number of move counter.

	1		Report the distance counter in meter.
	2		Report the no load (crash) counter.
	5		Report the clearable move counter.
Error	(3) Invalid operand.		

RDZ *Report Diagnostic Functions for the Z-Axis:*

RDZ Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2, 5	0	Reports selected diagnose function.
Response	***if***		
Selector	0		Report the number of move counter.
	1		Report the distance counter in meter.
	2		Report the no load (crash) counter.
	5		Report the clearable move counter.
Error	(3) Invalid operand.		

RDR *Report Diagnostic Functions for the R-Axis:*

RDR Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2, 5	0	Reports selected diagnose function.
Response	***if***		
Selector	0		Report the number of move counter.
	1		Report the distance counter in k°.
	2		Report the no load (crash) counter.
	5		Report the clearable move counter.
Error	(3) Invalid operand.		

RDG *Report Diagnostic Functions for the G-Axis:*

RDG Selector			
All the diagnostic values will be saved at power off of the instrument or if a power fail situation occurs. The stored data can be reported by this command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0..3, 5	0	Reports selected diagnose function.
Response	***if***		
Selector	0		Report the number of move counter.
	1		Report the distance counter in meter.
	2		Report the no load (crash) counter.
	3		Report the number of gripped plates.
	5		Report the clearable move counter.
Error	(3) Invalid operand.		

RIB *Report Baud Rate of BCS:*

RIB			
This command reports the baud rate of the communication between the ROMA to the Barcode Scanner.			
Parameter	***Validity***	***Default***	***Description***
None			
Response	***if***		
			Report the Baud Rate in bps.

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Error	(5) Device not implemented, (11) Sub device not ready yet

REB *Report error status of BCS*

REB

This command reports the error status of the barcode scanner. This information will be updated after each **TRB, ARC** or **BSM** command.

Parameter	***Validity***	***Default***	***Description***
None			

Response ***if***

Report the error status code of the last TRB, ARC or BSM command.

Response Code	**Cause**
0	Command not defined
1	Invalid command format
2	Operand not applicable
3	Operand ‘m’ is invalid
4	Wrong sequence of commands
5	Number in the command invalid
6	The operand “hhh…” is invalid
7	Operand not 0 or 1
8	Operand ‘n’ is invalid
9	Operand “nn” or “nnn” is invalid
10	Wrong data format of Hex operand
11	Value h’FF is invalid for “hhh…”
12	Operand too many characters
13	Operand “aaa” is invalid
14	Data in EEPROM may are damaged
15	Init setting data area defect
17	Invalid character sequence after ‘\’
18	Too many ‘!’ characters
99	Malfunction of the scanner
252	BCS in Set Mode[1]
253	Reference CODE 128 not set[1]
254	BCS in beam on mode[1]
255	No Error[1]

For further details see the correlatively command set of the barcode scanner: BL-600 user’s manual by Keyence.

Error	(5) Device not implemented, (11) Sub device not ready yet, (21) BCS communication error,

[1] Tecan defined errors

RMD *Report Multiple Download:*

RMD Layer

This command reports the download state of the Subdevices.
Answerformat:
<SD0><SD1><SD2><SD3><SD4><SD5><SD6><SD7><SD8><SD9><SD10><SD11><SD12><SD13><SD14><SD15>
To visualize the error codes as ASCII characters, an offset of 40 hexadecimal is added to the error value.
The download state ‘Y’ can be reset only by <SMD-1>. The download state data will not be remained at powerfail.

Parameter	***Validity***	***Default***	***Description***

Layer	0..7	0	Selects the communication layer of the L-CAN to be reported
Response	***if***		
			Report the download state of the subdevices @ = No download Error E = Device not implemented Y = Download Error, download for the Subdevice aborted Z = Download Active, Subdevice intended to be or being downloaded
Error	(3) Invalid operand.		

4.3 Set Commands (Base Functionality)

SFV Set Serial Number:

SFV Key, Number			
This command sets the device serial number. A total of sixteen characters can be stored. If the given string is longer then sixteen characters or the first character is not a number, an invalid operand error will be generated. This data will be stored definitely in non-volatile memory (Flash-EPROM) either when power down the instrument or while using the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Key	xxxxx		Secret.
Number	Up to 16 ASCII characters	0 ASCII	Serial number value stored as string.
Error	(3) Invalid operand.		

SGD Set Global Data:

SGD Selector, Value			
This command writes global data into an array in volatile memory. A maximum of 100 variables (type of integer) can be handled. This data will be stored definitely in non-volatile memory (Flash-EPROM) either when power down the instrument or while using the **AWE** command. CAUTION! This command is for internal use only (TECAN software). Overwriting of such variables can cause malfunction of the instrument and software.			
Parameter	***Validity***	***Default***	***Description***
Selector	1..100		Select the global data entry.
Value	-32768..32767	0	Global data value of signed integer size.
Error	(3) Invalid operand.		

SED Set FLASH-EPROM Data:

SED Selector, Value			
This command writes Flash-EPROM data into the volatile memory mirror. This data will be stored definitely in non-volatile memory (Flash-EPROM) either when power down the instrument or while using the **AWE** command. The size of this data array has not a fixed length. It depends on the actual loaded firmware version. The actual length can be reported by the **RED0** command. CAUTION! This command is for service purposes only (board exchange). Overwriting of such variables can cause malfunction of the instrument and software.			
Parameter	***Validity***	***Default***	***Description***
Selector	1..**RED0**		Select the Flash-EPROM data entry.
Value	-32768..32767	0	Flash-EPROM data value of signed integer size.
Error	(3) Invalid operand.		

SCC Send Broadcast Message to specific Slave Device:

SCC SlaveId, Cmd
This command sends messages with the type broadcast to specific devices connected to the CAN bus. Normally broadcast messages will be distributed to all participants of a network. With this command, it is possible to use the predefined broadcast commands and send them to only one specific participant in the network. If real broadcasts are required, the **SBC** command must be used instead of the **SCC** command. The specific supported broadcast command functions must be described in the command set of the target device. CAUTION: This command should be used only by trained personal.

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Parameter	Validity	Default	Description
SlaveId	0..127	0	Select the specific target node.
Cmd	0..255	0	Broadcast command.
Error	(3) Invalid operand.		

SBC *Send Broadcast Message to all Slave Devices:*

SBC Cmd			
This command sends messages with the type broadcast to all devices connected to the CAN bus. The specific supported broadcast command functions must be described in the command set of the target device. CAUTION: This command should be used only by trained personal.			
Parameter	***Validity***	***Default***	***Description***
Cmd	0..255	0	Broadcast command.
Error	(3) Invalid operand.		

SRS *Set Reset System:*

SRS			
This command resets the whole subsystem. All participants connected to the CAN bus will be reset and also this controlling DCU will perform a soft reset. After such a command, the system should be reconstructed and it should be in an initial state as like after a normal power up. CAUTION: This command should be used only by trained personal.			
Parameter	***Validity***	***Default***	***Description***
Error	None.		

SOF *Firmware Download:*

SOF Firmware			
This command initiates the firmware downloading process. The **SOF** command should not be used as string command itself. Only in connection with a download utility program, this command makes sense.			
Parameter	***Validity***	***Default***	***Description***
Firmware			[INTEL.HEX file format]
Error	None.		

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4.4 Set Commands (ROMA Specific Functionality)

SHW Set ROMA Hardware Version:

SHW Selector*			
This command sets the ROMA hardware version. The following combination can be installed and stored directly in non-volatile memory. The default outrigger distance and the default rotator displacement will be set by this command automatically. After changing the hardware type, the rotator axis must be reinitialized.			
Parameter	***Validity***	***Default***	***Description***
Selector*	0	0	Undefined ROMA type, no **PIA** is possible. Outrigger distance is 0.
	2		ROMA eccentric gripper type CW (right side orientation). Rotator initialization position is 0 and typical range is 0..2700. Outrigger distance is 800 and rotator displacement is 0.
	3		ROMA eccentric gripper type CCW (left side orientation). Rotator initialization position is 3600 and typical range is 900..3600. Outrigger distance is 800 and rotator displacement is 900.
	4		ROMA centric gripper type CW (right side orientation). Rotator initialization position is 0 and typical range is 0..2700. Outrigger distance is 0 and rotator displacement is 0.
	5		ROMA centric gripper type CCW (left side orientation). Rotator initialization position is 3600 and typical range is 900..3600. Outrigger distance is 0 and rotator displacement is 900.
Error	(3) Invalid operand.		

SOD Set Gripper Outrigger Distance:

SOD Distance*			
This command can be used to define an individual gripper outrigger distance (length of gripper). The default outrigger distance depends on the selected hardware set by the **SHW** command. The distance parameter can be stored in non-volatile memory (Flash-EPROM) with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Distance*	0..32767	0 or 800	Set the gripper outrigger distance.
Error	(3) Invalid operand.		

SAA Set Vector Coordinate Position into Table:

SAA Vector, X*, Y*, Z*, R*, G*, SpeedSel, ParamClass, WindowClass			
This command defines the vector coordinate positions where the ROMA should drive. There are up to 100 vectors storable. Every axis can be selected individually. If an axis is not defined (parameter not set), the axis will not move. If the vector definition is finished, it can be reported by the **RAA** command or it can be started by the **AAA** or **AAC** command. To define the ParamClasses or the WindowClasses us the **SPC** and the **STW** or the **SPT** command.			
Parameter	***Validity***	***Default***	***Description***
Vector	1..100		Selected the vector to be defined.
X*	Lower Limit … Upper Limit		Set absolute position in $^{1}/_{10}$ mm for the X-axis.
Y*	Lower Limit … Upper Limit		Set absolute position in $^{1}/_{10}$ mm for the Y-axis.
Z*	Lower Limit … Upper Limit		Set absolute position in $^{1}/_{10}$ mm for the Z-axis.

R*	Lower Limit … Upper Limit		Set absolute position in $^{1}/_{10}$ dg for the R-axis.
G*	Center Limit … Outer Limit		Set absolute position in $^{1}/_{10}$ mm for the G-axis.
SpeedSel	0..1	0	0 = slow speed, 1 = fast speed.
ParamClass	0..100	0	Sets the drive parameter class 0 = Common parameters set by SFX/Y/Z/R/G and SSX/Y/Z/R/G are used. (for AAC only)
WindowClass	0..100	0	Sets the Target Window Class 0 = **<SPT>** cmd (for AAC only)
Error	(3) Invalid operand, (5) device not implemented.		

SPC Set Drive Parameter Classes for AAC:

SPC Class, ParamNo, Param X*, Param Y*, Param Z*, Param R*, Param G*			
This command sets drive parameters for the **AAC** command. These paramters are stored in classes that can be assigned to the according move of the **AAC** command by the **SAA** command. These classes can be reported by the **RPC** command.			
Parameter	***Validity***	***Default***	***Description***
Class	1..100		Class in which the drive parameter will be stored
ParamNo	0…3		Parameter selector
0 Accel. ramp up	190..10'000	0	**Param X**: Acceleration X-axis in $^{10}/mm_{/s^2}$
	320.. 9'500	0	**Param Y**: Acceleration Y-axis in $^{10}/mm_{/s^2}$
	100.. 2'500	0	**Param Z**: Acceleration Z-axis in $^{10}/mm_{/s^2}$
	100..19'000	0	**Param R**: Acceleration R-axis in $^{10}/dg_{/s^2}$
	130.. 4'000	0	**Param G**: Acceleration G-axis in $^{10}/mm_{/s^2}$
1 Accel. ramp down	190..10'000	0	**Param X**: Deceleration X-axis in $^{10}/mm_{/s^2}$
	320.. 9'500	0	**Param Y**: Deceleration Y-axis in $^{10}/mm_{/s^2}$
	100.. 2'500	0	**Param Z**: Deceleration Z-axis in $^{10}/mm_{/s^2}$
	100..19'000	0	**Param R**: Deceleration R-axis in $^{10}/dg_{/s^2}$
	130.. 4'000	0	**Param G**: Deceleration G-axis in $^{10}/mm_{/s^2}$
2 SlowSpeed	1..4'000	0	**Param X**: Slow Speed X-axis
	1..4'000	0	**Param Y**: Slow Speed Y-axis
	1..2'500	0	**Param Z**: Slow Speed Z-axis
	1..4'000	0	**Param R**: Slow Speed R-axis
	1..4'000	0	**Param G**: Slow Speed G-axis
3 FastSpeed	50..10'000	0	**Param X**: Fast Speed X axis
	50..9'500	0	**Param Y**: Fast Speed Y-axis
	50..2'500	0	**Param Z**: Fast Speed Z-axis
	50..19'000	0	**Param R**: Fast Speed R-axis
	50..4'000	0	**Param G**: Fast Speed G-axis
Error	(3) Invalid operand.		

STW Set Target Window Classes for AAC:

STW Class, Window X*, Window Y*, Window Z*, Window R*, Window G*			
This command sets drive parameters for the **AAC** command. These parameters are stored in classes that can be assigned to the according move of the **AAC** command by the **SAA** command. These classes can be reported by the **RTW** command.			
Parameter	***Validity***	***Default***	***Description***
Class	1..100		Class in which the target window parameter will be stored
Window X*	0…5000	0	Set Target Window for X Axis in $^{1}/_{10}$ mm
Window Y*	0…5000	0	Set Target Window for Y Axis in $^{1}/_{10}$ mm
Window Z*	0…5000	0	Set Target Window for Z Axis in $^{1}/_{10}$ mm
Window R*	0…5000	0	Set Target Window for R Axis in $^{1}/_{10}$ °
Window G*	0…5000	0	Set Target Window for G Axis in $^{1}/_{10}$ mm
Error	(3) Invalid operand.		

SPT Set Permanent Target Window:

SPT WinX, WinY, WinZ, WinR, WinG			
This command sets the target window for the axes for all move commands. This Values can be stored permanently by AWE command.			
Parameter	***Validity***	***Default***	***Description***
WinX	0..5000	0	Set permanent Target Window for X Axis in $^{1}/_{10}$ mm
WinY	0..5000	0	Set permanent Target Window for Y Axis in $^{1}/_{10}$ mm
WinZ	0..5000	0	Set permanent Target Window for Z Axis in $^{1}/_{10}$ mm
WinR	0..5000	0	Set permanent Target Window for R Axis in $^{1}/_{10}$ °
WinG	0..5000	0	Set permanent Target Window for G Axis in $^{1}/_{10}$ mm
Error	(3) Invalid operand.		

SRA Remove Vector Coordinate Position from Table:

SRA			
This command is used to remove the whole vector table set by the **SAA** command.			
Parameter	***Validity***	***Default***	***Description***
Error	None.		

SRX Set X-Axis Range:

SRX Range*			
This command sets the absolute X-axis range. It can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Range*	0..1'000'000 and must be > X-Displace or > 0	1000	Set X-axis range in 1/10 mm.
Error	(3) Invalid operand.		

SRY Set Y-Axis Range:

SRY Range*			
This command sets the absolute Y-axis range. It can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Range*	0..1'000'000 and must be > Y-Displace or > 0	1000	Set Y-axis range in 1/10 mm.
Error	(3) Invalid operand.		

SRZ Set Z-Axis Range:

SRZ Range*			
This command sets the absolute Z-axis range in positive direction. Positive direction is above table. The negative range underneath the table can be defined by the **SOZ,xx** command. The **SRZ** range can be stored in non-volatile memory with the **AWE** command. After changing this parameter, this axis must be reinitialized again.			
Parameter	***Validity***	***Default***	***Description***
Range*	0..1'000'000	1000	Set Z-axis range in 1/10 mm.

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Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.

SRR *Set R-Axis Range:*

SRR Range*			
This command sets the absolute R-axis range. It can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Range*	0..1'000'000	1000	Set R-axis range in 1/10 dg.
Error	(3) Invalid operand.		

SRG *Set G-Axis Range:*

SRG Range*			
This command sets the absolute G-axis range. It can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Range*	0..1'000'000 and must be > G-Displace	1560	Set G-axis range in 1/10 mm.
Error	(3) Invalid operand.		

SOX *Set Initialization Offset for X-Axis:*

SOX InitOffset*, Displace*			
This command sets the initialization offset and the displacement offset for the X-axis to the entered values. Both parameter entered are in action immediately after this command is given. They can be stored in non-volatile memory by the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
InitOffset*	0..X-Range + X-Offset – X-Displace or 32767	10	Set X-axis initialization offset in 1/10 mm.
Displace*	-250..32767	0	Set X-axis displacement offset in 1/10 mm.
Error	(3) Invalid operand.		

SOY *Set Initialization Offset for Y-Axis:*

SOY InitOffset*, Displace*			
This command sets the initialization offset and the displacement offset for the Y-axis to the entered values. Both parameter entered are in action immediately after this command is given. They can be stored in non-volatile memory by the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
InitOffset*	10..Y-Range + Y-Offset – Y-Displace or 32767	10	Set Y-axis initialization offset in 1/10 mm.
Displace*	-1000..32767	-620	Set Y-axis displacement offset in 1/10 mm.
Error	(3) Invalid operand.		

SOZ *Set Initialization Offset for Z-Axis:*

SOZ InitOffset*, NegRange*			
This command sets the initialization offset and the negative axis range for the Z-axis to the entered values. Both parameter entered are in action immediately after this command is given. They can be stored in non-volatile memory by the **AWE** command. After changing one or both of the parameters, this axis must be reinitialized again.			
Parameter	***Validity***	***Default***	***Description***
InitOffset*	10..Z-Range + Z-Offset or 32767	10	Set Z-axis initialization offset in 1/10 mm.
NegRange*	-32768..0	0	Set Z-axis negative range in 1/10 mm.

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Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.

SOR Set Initialization Offset for R-Axis:

SOR InitOffset*			
This command sets the initialization offset for the R-axis to the entered value. This parameters is in action immediately after this command is given. It can be stored in non-volatile memory by the **AWE** command. The displacement of this axis will be defined by the **SHW** setting, which gives a constant value to this parameter.			
Parameter	***Validity***	***Default***	***Description***
InitOffset*	10..R-Range + R-Offset or 32767	50	Set R-axis initialization offset in 1/10 dg.
Error	(3) Invalid operand.		

SOG Set Initialization Offset for G-Axis:

SOG InitOffset*, Displace*			
This command sets the initialization offset and the displacement offset between the gripper pins (inner distance) to the G-axis. Both parameters are in action immediately after this command is given. They can be stored in non-volatile memory by the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
InitOffset*	10..G-Range + G-Offset – G-Displace or 32767	20	Set G-axis initialization offset in 1/10 mm.
Displace*	0..32767	560	Set G-axis displacement offset in 1/10 mm.
Error	(3) Invalid operand.		

SSX Set Slow Speed for X-Axis:

SSX Speed*			
This function sets the slow speed for the X-axis. This setting is in action while using **AAA**, **MAA**, **MRX** or **MAX** commands without the SlowSpeed selector. It is possible to change the speed on the fly, by using this command via the alternative access channel.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..4000	1000	Set X-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SSY Set Slow Speed for Y-Axis:

SSY Speed*			
This function sets the slow speed for the Y-axis. This setting is in action while using **AAA**, **MAA**, and **MRY** or **MAY** commands without the SlowSpeed selector. It is possible to change the speed on the fly, by using this command via the alternative access channel.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..4000	800	Set Y-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SSZ Set Slow Speed for Z-Axis:

SSZ Speed*			
This function sets the slow speed for the Z-axis. This setting is in action while using **AAA**, **MAA**, **MRZ** or **MAZ** commands without the SlowSpeed selector. It is possible to change the speed on the fly, by using this command via the alternative access channel.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..2500	500	Set Z-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

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SSR *Set Slow Speed for R-Axis:*

SSR Speed*			
This function sets the slow speed for the R-axis. This setting is in action while using **AAA**, **MAA**, **MRR** or **MAR** commands without the SlowSpeed selector. It is possible to change the speed on the fly, by using this command via the alternative access channel.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..4000	600	Set R-axis slow speed in 1/10 dg/s.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SSG *Set Slow Speed for G-Axis:*

SSG Speed*			
This function sets the slow speed for the G-axis. This setting is in action while using **AAA**, **MAA**, **MRG** or **MAG** commands without the SlowSpeed selector. It is possible to change the speed on the fly, by using this command via the alternative access channel.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..4000	100	Set G-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SFX *Set Fast Speed and Acceleration for X-Axis:*

SFX Speed*, Accel*			
This command sets the X-axis fast speed and acceleration. These settings are in use while using **AAA, AAC, PAA**, **PRX** or **PAX** commands. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	50..10000	10000	Set X-axis fast speed in 1/10 mm/s.
Accel*	190..10000	1600	Set X-axis acceleration in 1/10 mm/s^2.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SFY *Set Fast Speed and Acceleration for Y-Axis:*

SFY Speed*, Accel*			
This command sets the Y-axis fast speed and acceleration. These settings are in use while using **AAA**, **AAC, PAA**, **PRY** or **PAY** commands. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	50..9500	3500	Set Y-axis fast speed in 1/10 mm/s.
Accel*	320..9500	1000	Set Y-axis acceleration in 1/10 mm/s^2.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SFZ *Set Fast Speed and Acceleration for Z-Axis:*

SFZ Speed*, Accel*			
This command sets the Z-axis fast speed and acceleration. These settings are in use while using **AAA**, **AAC, PAA, PRZ** or **PAZ** commands. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	50..2500	1300	Set Z-axis fast speed in 1/10 mm/s.
Accel*	100..2500	850	Set Z-axis acceleration in 1/10 mm/s^2.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

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SFR Set Fast Speed and Acceleration for R-Axis:

SFR Speed*, Accel*			
This command sets the R-axis fast speed and acceleration. These settings are in use while using **AAA**, **PAA**, **PRR** or **PAR** commands. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	50..19000	2000	Set R-axis fast speed in 1/10 dg/s.
Accel*	100..19000	600	Set R-axis acceleration in 1/10 dg/s^2.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SFG Set Fast Speed and Acceleration for G-Axis:

SFG Speed*, Accel*			
This command sets the G-axis fast speed and acceleration. These settings are in use while using **AAA**, **PAA**, **PRG** or **PAG** commands. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	50..4000	400	Set G-axis fast speed in 1/10 mm/s.
Accel*	130..4000	300	Set G-axis acceleration in 1/10 mm/s^2.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SGG Set Gripper Parameters:

SGG Speed*, PWMLimit*, CurLimit*			
The **SGG** command sets the gripper speed, PWM (pulse width modulation) limit and the maximal allowed current (in volatile memory) that are used while gripping a plate with the **AGR** command. For detailed information about PWM limit and current limit, read the *DC Servo 2 Controller Command Summary* document. With the **AWE** command, the values can be stored in the non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
Speed*	1..1500	200	Set gripper search speed in 1/10 mm/s.
PWMLimit*	0..249	75	Set gripper PWMLimit in increments.
CurLimit*	0..3	0	Set gripper current.
Error	(3) Invalid operand.		

SAX Set Scale Adjust Factor for X-Axis:

SAX ScaleAdj*			
This command defines the scale adjust factor of the X-axis within a defined range. Default scale adjust is 10000. This value can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
ScaleAdj*	9500..10500	10000	Set X-axis scale adjust factor where 10000 is 100%.
Error	(3) Invalid operand.		

SAY Set Scale Adjust Factor for Y-Axis:

SAY ScaleAdj*			
This command defines the scale adjust factor of the Y-axis within a defined range. Default scale adjust is 10000. This value can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
ScaleAdj*	9500..10500	10000	Set Y-axis scale adjust factor where 10000 is 100%.
Error	(3) Invalid operand.		

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SAZ Set Scale Adjust Factor for Z-Axis:

SAZ ScaleAdj*			
This command defines the scale adjust factor of the Z-axis within a defined range. Default scale adjust is 10000. This value can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
ScaleAdj*	9500..10500	10000	Set Z-axis scale adjust factor where 10000 is 100%.
Error	(3) Invalid operand.		

SAR Set Scale Adjust Factor for R-Axis:

SAR ScaleAdj*			
This command defines the scale adjust factor of the R-axis within a defined range. Default scale adjust is 10000. This value can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
ScaleAdj*	9500..10500	10000	Set R-axis scale adjust factor where 10000 is 100%.
Error	(3) Invalid operand.		

SAG Set Scale Adjust Factor for G-Axis:

SAG ScaleAdj*			
This command defines the scale adjust factor of the G-axis within a defined range. Default scale adjust is 10000. This value can be stored in non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
ScaleAdj*	9500..10500	10000	Set G-axis scale adjust factor where 10000 is 100%.
Error	(3) Invalid operand.		

SLX Set X-Axis Scaling Factor and Motion Direction:

SLX ScaleFact*			
CAUTION! This command should never be used in the normal operation with an instrument. Wrong definition causes malfunction and could destroy the instrument! With this command, it's possible to redefine the X-axis scale factor. In some special applications it's maybe useful to redefine this value. Inverting the sign of the scale factor changes the motor rotation direction. This value can't be stored in non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
ScaleFact*	-32768..32767	-1250	Set X-axis scale factor.
Error	(3) Invalid operand.		

SLY Set Y-Axis Scaling Factor and Motion Direction:

SLY ScaleFact*			
CAUTION! This command should never be used in the normal operation with an instrument. Wrong definition causes malfunction and could destroy the instrument! With this command, it's possible to redefine the Y-axis scale factor. In some special applications it's maybe useful to redefine this value. Inverting the sign of the scale factor changes the motor rotation direction. This value can't be stored in non-volatile memory.			
Parameter	***Validity***	***Default***	***Description***
ScaleFact*	-32768..32767	1037	Set Y-axis scale factor.
Error	(3) Invalid operand.		

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SLZ Set Z-Axis Scaling Factor and Motion Direction:

SLZ ScaleFact*

CAUTION! This command should never be used in the normal operation with an instrument. Wrong definition causes malfunction and could destroy the instrument! With this command, it's possible to redefine the Z-axis scale factor. In some special applications it's maybe useful to redefine this value. Inverting the sign of the scale factor changes the motor rotation direction. This value can't be stored in non-volatile memory.

Parameter	*Validity*	*Default*	*Description*
ScaleFact*	-32768..32767	283	Set Z-axis scale factor.
Error	(3) Invalid operand.		

SLR Set R-Axis Scaling Factor and Motion Direction:

SLR ScaleFact*

CAUTION! This command should never be used in the normal operation with an instrument. Wrong definition causes malfunction and could destroy the instrument! With this command, it's possible to redefine the R-axis scale factor. In some special applications it's maybe useful to redefine this value. Inverting the sign of the scale factor changes the motor rotation direction. This value can't be stored in non-volatile memory.

Parameter	*Validity*	*Default*	*Description*
ScaleFact*	-32768..32767	2045	Set R-axis scale factor.
Error	(3) Invalid operand.		

SLG Set G-Axis Scaling Factor and Motion Direction:

SLG ScaleFact*

CAUTION! This command should never be used in the normal operation with an instrument. Wrong definition causes malfunction and could destroy the instrument! With this command, it's possible to redefine the G-axis scale factor. In some special applications it's maybe useful to redefine this value. Inverting the sign of the scale factor changes the motor rotation direction. This value can't be stored in non-volatile memory.

Parameter	*Validity*	*Default*	*Description*
ScaleFact*	-32768..32767	820	Set G-axis scale factor.
Error	(3) Invalid operand.		

SSM Set X-Axis Smooth Move Mode:

SSM Mode*, AxisHarm*

This command sets the X-axis smooth move and the Y..G axis harmonization mode. By default the smooth move and the Y..G axis harmonization mode are active.

X-axis smooth move : The acceleration and speed parameters will be recalculated depending on the distance to move. So a smooth motion with the same duration can be driven.

Y..G Axis harmonization: The speed of the Y, Z, R and G axes will be recalculated to achieve a collective arrival of these axes.

When selecting mode 1, the normal acceleration and speed parameters set by the **SFX/Y/Z/R/G** command are in use. The mode parameters can be stored in non-volatile memory (Flash-EPROM) with the **AWE** command.

Parameter	*Validity*	*Default*	*Description*
Mode*	0..1		0 = Smooth move active 1 = Normal acceleration and speed parameter set by the **SFX** command are in use.
AxisHarm*	0..1		0 = Harmonisation of Y..G axis is active 1 = Normal acceleration and speed parameter set by the

			SFY/Z/R/G command are in use.
Error	(3) Invalid operand.		

SBK *Set Z-Axis Break On or Off:*

SBK OnOff*			
This command can be used to turn on or off the Z-axis break. In normal operation, the break will be controlled by the moving commands. There is no need to control the break manually, except for service purposes.			
Parameter	***Validity***	***Default***	***Description***
OnOff*	0..1		0 = break active (power off), 1 = break released (power on).
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

SMC *Set all Clearable ROMA Axis Move Counters to Zero:*

SMC Key			
This command sets the entire ROMA move counter, reportable with the **RDX/Y/Z/R/G5** command to zero. The value of the counter that can be reported by the **RDX/Y/Z/R/G0** command remains at the same value as before.			
Parameter	***Validity***	***Default***	***Description***
Key	xxxxx		Secret.
Error	(3) Invalid operand.		

SMD *Set Multiple Download:*

SMD Layer, SubDev-Set			
This command sets the subdevices to be downloaded by the next multiple download **MMD**.			
Parameter	***Validity***	***Default***	***Description***
Layer	-1..7	0	-1 = Reset all subdevices in all layers. 0..7 = Select the communicaton layer of the L-CAN.
SubDev-Set	0..65'535	0	Set the bit set of the subdevices to be downloaded
Error	(3) Invalid operand		

4.5 Action Commands (Base Functionality)

AWE *Write System Parameters to FLASH-EEPROM:*

AWE			
This command writes all parameters set by the following commands to the Flash-EEPROM (non-volatile memory). **SRX/Y/Z/R/G** absolute range. **SOX/Y/Z/R/G** initialization offset and displacement. **SFX/Y/Z/R/G** fast speed and acceleration. **SAX/Y/Z/R/G** scale adjusts. **SHW** Hardware Version **SGG** gripper parameter. **SOD** outrigger distance. **SSM** smooth move mode. **SFV** Set Serial Number. **SGD** Set Global Data. **SED** Set FLASH-EPROM Data.			
Parameter	***Validity***	***Default***	***Description***
Error	(13) No access to serial EEPROM.		

ARE *Read System Parameters to FLASH-EPROM:*

ARE			
This command reads all parameters set by the following commands from the Flash-EPROM (non-volatile memory) into the working memory (volatile memory). **SRX/Y/Z/R/G** absolute range. **SOX/Y/Z/R/G** initialization offset and displacement. **SFX/Y/Z/R/G** fast speed and acceleration. **SAX/Y/Z/R/G** scale adjusts. **SHW** Hardware Version **SGG** gripper parameter. **SOD** outrigger distance. **SSM** smooth move mode. **SFV** Set Serial Number. **SGD** Set Global Data. **SED** Set FLASH-EPROM Data.			
Parameter	***Validity***	***Default***	***Description***
Error	(13) No access to serial EEPROM.		

ACW *Erase all Setup Data in the Flash:*

ACW Key			
Erases all data in the Flash this includes Diagnose- and Parameter Data. The default data will be stored. NOTE! This command is for service purposes only. CAUTION! Inappropriate usage of this command causes mismatched reading results!			
Parameter	***Validity***	***Default***	***Description***
Key	xxxxx		Secret.
Error	(3) Invalid operand.		

Txx *Pipeline to Slave Device:*

T x y Cmd			
Send a command transparent to a sub-device. <u>Example:</u> C1T31RFV2 → Send the command RFV2 (Report Serial number) to the Subdevice T31. The command goes to O1 of the local CAN of C1 and executes the required command corresponding the command set of the subdevice.			
Parameter	***Validity***	***Default***	***Description***
x	0..7	0	Layer number coded in alpha characters (A,D, O , Q,..)
y	0..15	0	Device number, HEX format
Cmd	Depends on CS		Command of sub-device
Error	(3) Invalid operand, Sub-Device Specific Errors		

4.6 Action Commands (ROMA Specific Functionality)

PIA Position Initialization X/Y/Z/R/G – Axis:

PIA Selector			
This is the command, to initialize the ROMA X/Y/Z/Rotator and Gripper-axis. No movement command for the ROMA will be accepted before the **PIA** or an axis specific initialization command is executed. When the **PIA** command is entered, all drives try to move to their initialization positions with a default init speed. With the selector it is possible to disable the X-axis. This is useful after a pre initialization with **PIB** command. When the init position of each axis has been detected, they move to an initial offset position. It is possible that the ROMA can't find a free position to initialize all drives, in this case the ROMA must be moved to a free location manually. The initialization procedure depends on the selected hardware defined by the **SHW** command.			
Parameter	***Validity***	***Default***	***Description***
Selector	0	0	Normal initialization.
	1		X-axis disabled.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (14) HW not defined, (30) Invalid servo version.		

PIB Position Pre Initialization:

PIB			
The **PIB** command is used to perform a pre initialization of the arm. This must be to prevent a collision with an obstacle in the rear part of the instrument while initialization. After performing the **PIB**, normally a **PIA** without X-axis will be performed. The goal of this command is to bring the Y-axis to the front of the instrument. After doing a **PIB**, all axis have a not initialized status. The initialization procedure depends on the selected hardware defined by the **SHW** command.			
Parameter	***Validity***	***Default***	***Description***
Error	(1) Initialization failed, (5) Device not implemented, (14) HW not defined.		

PIX Position Initialization X-Axis:

PIX InitSpeed*			
This function initializes only the X-axis of the ROMA. An override motor speed for axis initialization may be placed in the InitSpeed parameter. A command without any parameter uses the default speed to initialize.			
Parameter	***Validity***	***Default***	***Description***
InitSpeed*	5..1500	400	Set individual initialization speed in 1/10 mm/s.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (30) Invalid servo version.		

PIY Position Initialization Y-Axis:

PIY InitSpeed*			
This function initializes only the Y-axis of the ROMA. An override motor speed for axis initialization may be placed in the InitSpeed parameter. A command without any parameter uses the default speed to initialize.			
Parameter	***Validity***	***Default***	***Description***
InitSpeed*	5..1500	350	Set individual initialization speed in 1/10 mm/s.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (30) Invalid servo version.		

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PIZ Position Initialization Z-Axis:

PIZ InitSpeed*			
This function initializes only the Z-axis of the ROMA. An override motor speed for axis initialization may be placed in the InitSpeed parameter. A command without any parameter uses the default speed to initialize. The init position of the Z-drive is the range position set by the **SRZ** command. That means, that the Z-zero position must be somewhere near the worktable.			
Parameter	***Validity***	***Default***	***Description***
InitSpeed*	5..1500	200	Set individual initialization speed in 1/10 mm/s.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (6) CAN answer timeout, (30) Invalid servo version.		

PIR Position Initialization R-Axis:

PIR InitSpeed*			
This function initializes only the R-axis of the ROMA. An override motor speed for axis initialization may be placed in the InitSpeed parameter. A command without any parameter uses the default speed to initialize.			
Parameter	***Validity***	***Default***	***Description***
InitSpeed*	5..1500	500	Set individual initialization speed in 1/10 dg/s.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (30) Invalid servo version.		

PIG Position Initialization G-Axis:

PIG InitSpeed*			
This function initializes only the G-axis of the ROMA. An override motor speed for axis initialization may be placed in the InitSpeed parameter. A command without any parameter uses the default speed to initialize.			
Parameter	***Validity***	***Default***	***Description***
InitSpeed*	5..1500	200	Set individual initialization speed in 1/10 mm/s.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (30) Invalid servo version.		

PIF Fake Initialization X/Y/Z/R/G – Axis:

PIF			
The **PIF** command tells the firmware that all axis have been initialized, however, no mechanical initialization is executed. The current position coordinate of each axis is still valid, except if it is not initialized. In case of a not initialized axis, the position will be set to zero or maximal range. CAUTION: Incorrect usage of this command may damage the instrument.			
Parameter	***Validity***	***Default***	***Description***
Error	None.		

PAX Position Absolute X-Axis:

PAX X*			
Moves the X-axis to an absolute position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFX** command.			
Parameter	***Validity***	***Default***	***Description***
X*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

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PAY *Position Absolute Y-Axis:*

PAY Y*			
Moves the Y-axis to an absolute position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFY** command.			
Parameter	***Validity***	***Default***	***Description***
Y*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PAZ *Position Absolute Z-Axis:*

PAZ Z*			
Moves the Z-axis to an absolute position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFZ** command.			
Parameter	***Validity***	***Default***	***Description***
Z*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PAR *Position Absolute R-Axis:*

PAR R*			
Rotates the R-axis to an absolute position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFR** command.			
Parameter	***Validity***	***Default***	***Description***
R*	Lower Limit … Upper Limit	0	Absolute position in 1/10 dg.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PAG *Position Absolute G-Axis:*

PAG G*			
Moves the G-axis to an absolute position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFG** command.			
Parameter	***Validity***	***Default***	***Description***
G*	Center Limit … Outer Limit	0	Absolute position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PRX *Position Relative X-Axis:*

PRX +/-X			
Moves the X-axis relative to its actual position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFX** command.			
Parameter	***Validity***	***Default***	***Description***
+/-X	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

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PRY Position Relative Y-Axis:

PRY +/-Y			
Moves the Y-axis relative to its actual position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFY** command.			
Parameter	***Validity***	***Default***	***Description***
+/-Y	(-InitOffset + 10 + Y-Displace)..Y-Range	0	Relative position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PRZ Position Relative Z-Axis:

PRZ +/-Z			
Moves the Z-axis relative to its actual position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFZ** command.			
Parameter	***Validity***	***Default***	***Description***
+/-Z	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PRR Position Relative R-Axis:

PRR +/-R			
Rotates the R-axis relative to its actual position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the **SFR** command.			
Parameter	***Validity***	***Default***	***Description***
+/-R	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PRG Position Relative G-Axis:

PRG +/-G			
Moves the G-axis relative to its actual position, leaving the other axis positions unchanged. The movement will be performed with the speed and acceleration setting done by the SFG command.			
Parameter	***Validity***	***Default***	***Description***
+/-G	Target between Center Limit … Outer Limit	0	Relative position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MAX Position Absolute with Slow Speed X-Axis:

MAX X*, Speed*			
Moves the X-axis with selected slow speed to an absolute position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSX** command is used.			
Parameter	***Validity***	***Default***	***Description***
X*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Speed*	1..4000	1000	Set X-axis slow speed in 1/10 mm/s.

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Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.

MAY *Position Absolute with Slow Speed Y-Axis:*

MAY Y*, Speed*			
Moves the Y-axis with selected slow speed to an absolute position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSY** command is used.			
Parameter	***Validity***	***Default***	***Description***
Y*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Speed*	1..4000	800	Set Y-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MAZ *Position Absolute with Slow Speed Z-Axis:*

MAZ Z*, Speed*			
Moves the Z-axis with selected slow speed to an absolute position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSZ** command is used.			
Parameter	***Validity***	***Default***	***Description***
Z*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Speed*	1..4000	500	Set Z-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MAR *Position Absolute with Slow Speed R-Axis:*

MAR R*, Speed*			
Rotates the R-axis with selected slow speed to an absolute position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSR** command is used.			
Parameter	***Validity***	***Default***	***Description***
R*	Lower Limit … Upper Limit	0	Absolute position in 1/10 mm.
Speed*	1..4000	600	Set R-axis slow speed in 1/10 dg/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MAG *Position Absolute with Slow Speed G-Axis:*

MAG G*, Speed*			
Moves the G-axis with selected slow speed to an absolute position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSG** command is used.			
Parameter	***Validity***	***Default***	***Description***
G*	Center Limit … Outer Limit	0	Absolute position in 1/10 mm.
Speed*	1..4000	100	Set G-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MRX *Position Relative with Slow Speed X-Axis:*

MRX +/-X, Speed*			
Moves the X-axis with selected slow speed relative to its actual position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSX** command is used.			
Parameter	***Validity***	***Default***	***Description***
+/-X	Target between	0	Relative position in 1/10 mm.

	Lower Limit … Upper Limit		
Speed*	1..4000	1000	Set X-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MRY *Position Relative with Slow Speed Y-Axis:*

MRY +/-Y, Speed*			
Moves the Y-axis with selected slow speed relative to its actual position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSY** command is used.			
Parameter	***Validity***	***Default***	***Description***
+/-Y	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 mm.
Speed*	1..4000	800	Set Y-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MRZ *Position Relative with Slow Speed Z-Axis:*

MRZ +/-Z, Speed*			
Moves the Z-axis with selected slow speed relative to its actual position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSZ** command is used.			
Parameter	***Validity***	***Default***	***Description***
+/-Z	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 mm.
Speed*	1..4000	500	Set Z-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MRR *Position Relative with Slow Speed R-Axis:*

MRR +/-R, Speed*			
Rotates the R-axis with selected slow speed relative to its actual position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSR** command is used.			
Parameter	***Validity***	***Default***	***Description***
+/-R	Target between Lower Limit … Upper Limit	0	Relative position in 1/10 dg.
Speed*	1..4000	600	Set R-axis slow speed in 1/10 dg/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MRG *Position Relative with Slow Speed G-Axis:*

MRG +/-G, Speed*			
Moves the G-axis with selected slow speed relative to its actual position, leaving the other axis positions unchanged. If no speed is selected, the slow speed given by the **SSG** command is used.			
Parameter	***Validity***	***Default***	***Description***
+/-G	Target between Center Limit … Outer Limit	0	Relative position in 1/10 mm.
Speed*	1..4000	100	Set G-axis slow speed in 1/10 mm/s.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

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PAA *Position Absolute for all Axis:*

PAA X*, Y*, Z*, R*, G*			
The **PAA** command moves the X/Y/Z/R and G-axis to the entered positions in the absolute field. Movement will be performed with speed and acceleration settings, defined by the **SFX/Y/Z/R/G** commands. Axis that have no position parameter selected, do not perform any action.			
Parameter	***Validity***	***Default***	***Description***
X*	Lower Limit … Upper Limit		Set absolute position in 1/10 mm for the X-axis.
Y*	Lower Limit … Upper Limit		Set absolute position in 1/10 mm for the Y-axis.
Z*	Lower Limit … Upper Limit		Set absolute position in 1/10 mm for the Z-axis.
R*	Lower Limit … Upper Limit		Set absolute position in 1/10 dg for the R-axis.
G*	Center Limit … Outer Limit		Set absolute position in 1/10 mm for the G-axis.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

PRA *Position Relative for all Axis:*

PRA X, Y, Z, R, G			
The **PRA** command moves the X/Y/Z/R and G-axis relative to its actual position. Movement will be performed with speed and acceleration settings, defined by the **SFX/Y/Z/R/G** commands. Axis that have no position parameter selected, do not perform any action.			
Parameter	***Validity***	***Default***	***Description***
X	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the X-axis.
Y	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the Y-axis.
Z	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the Z-axis.
R	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 dg for the R-axis.
G	Target between Center Limit … Outer Limit	0	Set absolute position in 1/10 mm for the G-axis.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

MAA *Position Absolute with Slow Speed for all Axis:*

MAA X*, Y*, Z*, R*, G*			
The **MAA** command moves the X/Y/Z/R and G-axis to the entered positions in the absolute field. Movement will be performed with speed settings, defined by the **SSX/Y/Z/R/G** commands. Axis that have no position parameter selected, do not perform any action.			
Parameter	***Validity***	***Default***	***Description***
X*	Lower Limit … Upper Limit		Set absolute position in 1/10 mm for the X-axis.
Y*	Lower Limit … Upper Limit		Set absolute position in 1/10 mm for the Y-axis.
Z*	Lower Limit … Upper		Set absolute position in 1/10 mm for the Z-axis.

	Limit	
R*	Lower Limit … Upper Limit	Set absolute position in 1/10 dg for the R-axis.
G*	Center Limit … Outer Limit	Set absolute position in 1/10 mm for the G-axis.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.	

MRA *Position Relative with Slow Speed for all Axis:*

MRA X, Y, Z, R, G

The **MRA** command moves the X/Y/Z/R and G-axis relative to its actual position. Movement will be performed with speed settings, defined by the **SSX/Y/Z/R/G** commands. Axis that have no position parameter selected, do not perform any action.

Parameter	***Validity***	***Default***	***Description***
X	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the X-axis.
Y	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the Y-axis.
Z	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 mm for the Z-axis.
R	Target between Lower Limit … Upper Limit	0	Set absolute position in 1/10 dg for the R-axis.
G	Target between Center Limit … Outer Limit	0	Set absolute position in 1/10 mm for the G-axis.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

AAA *Action Move to Vector Coordinate Position:*

AAA

This command starts the coordinate movement, built by the coordinate table. The point where the *ROMA* should drive to, can be defined with the command **SAA**. There is a maximum of hundred table entries allowed. Axis, which aren't defined, do not perform any action. The movements are done with the selected speed settings. After successful movement to all position in the table, the coordinate table is empty again.

Parameter	***Validity***	***Default***	***Description***
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.		

AAC *Action Move to Vector Coordinate Position with class attributes:*

AAC

This command starts the coordinate movement, built by the coordinate table. The point where the *ROMA* should drive to, can be defined with the command **SAA**. There is a maximum of hundred table entries allowed. Axis, which aren't defined, do not perform any action. The movements are done with the selected speed class settings set by **SPC** and **STW**. After successful movement to all position in the table, the coordinate table is empty again.
CAUTION: The smooth move functionality is not active for this command

Parameter	***Validity***	***Default***	***Description***

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Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (10) drive no load.

AGR *Grip a Plate:*

AGR Pos

This command is gripping a plate with the G-axis at the current X/Y/Z/R-position. The plate must be fetched within the current position (start) and the absolute position (target) defined by the parameter Pos. If no plate can be fetched within the search distance a Plate not Fetched error will be generated. The speed, force and maximal allowed current to grip can be defined with the **SGG** command.

Parameter	***Validity***	***Default***	***Description***
Pos	(-InitOffset + 10 + G-Displace)..G-Range		Target position in 1/10 mm.
Error	(3) Invalid operand, (4) CAN ack problem, (5) Device not implemented, (7) Device not initialized, (9) plate not fetched.		

ARX *Auto Range X-Axis:*

ARX Scale, Offset*

This command is measuring the maximal reachable range of the X-axis. The scale adjust can be either switched to default or the actual setting can be used. Also an alternative initialization offset can be given. If the offset parameter is not used, the standard offset set by the **SOX** command will be used. The determined scale adjust value can be stored into the non-volatile memory with the **AWE** command.

Parameter	***Validity***	***Default***	***Description***
Scale	0..1	0	0 = use original scale adjust, 1 = reset scale adjust to 10000.
Offset*	0..InitOffset		Set alternative initialization offset in 1/10 mm.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented.		

ARY *Auto Range Y-Axis:*

ARY Scale, Offset*

This command is measuring the maximal reachable range of the Y-axis. The scale adjust can be either switched to default or the actual setting can be used. Also an alternative initialization offset can be given. If the offset parameter is not used, the standard offset set by the **SOY** command will be used. The determined scale adjust value can be stored into the non-volatile memory with the **AWE** command.

Parameter	***Validity***	***Default***	***Description***
Scale	0..1	0	0 = use original scale adjust, 1 = reset scale adjust to 10000.
Offset*	0..InitOffset		Set alternative initialization offset in 1/10 mm.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented.		

ARZ *Auto Range Z-Axis:*

ARZ Scale, Offset*

This command is measuring the maximal reachable range of the Z-axis. The scale adjust can be either switched to default or the actual setting can be used. Also an alternative initialization offset can be given. If the offset parameter is not used, the standard offset set by the **SOZ** command will be used. The determined scale adjust value can be stored into the non-volatile memory with the **AWE** command.

Parameter	***Validity***	***Default***	***Description***
Scale	0..1	0	0 = use original scale adjust, 1 = reset scale adjust to 10000.
Offset*	0..InitOffset		Set alternative initialization offset in 1/10 mm.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented.		

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ARR *Auto Range R-Axis:*

ARR Scale, Offset*			
This command is measuring the maximal reachable range of the R-axis. The scale adjust can be either switched to default or the actual setting can be used. Also an alternative initialization offset can be given. If the offset parameter is not used, the standard offset set by the **SOR** command will be used. The determined scale adjust value can be stored into the non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Scale	0..1	0	0 = use original scale adjust, 1 = reset scale adjust to 10000.
Offset*	0..InitOffset		Set alternative initialization offset in 1/10 mm.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented.		

ARG Auto Range G-Axis:

ARG Scale, Offset*			
This command is measuring the maximal reachable range of the G-axis. The scale adjust can be either switched to default or the actual setting can be used. Also an alternative initialization offset can be given. If the offset parameter is not used, the standard offset set by the **SOG** command will be used. The determined scale adjust value can be stored into the non-volatile memory with the **AWE** command.			
Parameter	***Validity***	***Default***	***Description***
Scale	0..1	0	0 = use original scale adjust, 1 = reset scale adjust to 10000.
Offset*	0..InitOffset		Set alternative initialization offset in 1/10 mm.
Error	(1) Initialization failed, (3) Invalid operand, (4) CAN ack problem, (5) Device not implemented.		

TRB Transfer to barcode scanner:

TRB Command			
This command can be used to transfer commands straight to the barcode scanner. The Error Codes of the BCS will be converted into Tecan standard numeric error codes, as described in chapter 4.9. Use this command to configure and test the scanner. The Sub device error (28) can be reported more detailed by the **REB** command. ATTENTION: Communication parameters of the BCS must not be changed by this command; otherwise the BCS will try to work with wrong communication settings. The BCS-command “RESET” as well changes the communication protocol to default and should not be used.			
Parameter	***Validity***	***Default***	***Description***
Command	ASCII characters according to the command set of the barcode scanner		*Example commands:* LON Switch on Barcode scanner and read Barcode. LOFF Switch off Barcode scanner. SSET Enter on Set Mode SEND Leave Set Mode For further commands see the correlatively command set of the barcode scanner: BL-600 user’s manual by Keyence.
Error	(3) Invalid operand, (5) Device not implemented, (11) Sub device not ready yet, (21) BCS communication error, (28) Sub device error.		

ABR Read barcode:

ABR TimeOut			
This command reads a barcode and reports it. The answer string for a non-readable barcode is ***. Barcodes up to 56 characters can be read.			
Parameter	***Validity***	***Default***	***Description***
TimeOut	0,100-3000	1000 (0)	Selects the time out to read a barcode in ms. 0 = 1000ms
Error	(3) Invalid operand, (5) Device not implemented, (11) Sub device not ready yet, (21) BCS communication error, (28) Sub device error.		

ARC Read reference barcode:

ARC TimeOut			
This command reads a reference barcode and verifies it. The verification barcode is a “CODE 128” and contains “RBCD”. For a successful verification the barcode type “CODE 128” has to be set. An error will be sent if the verification has failed. The Sub device error (28) can be reported more detailed by the **REB** command.			
Parameter	***Validity***	***Default***	***Description***
TimeOut	0,100-3000	0 (1000 ms)	Selects the time out to read a barcode in ms. 0 = 1000 ms
Error	(3) Invalid operand, (5) Device not implemented, (11) Sub device not ready yet, (17) Verification failed, (21) BCS communication error, (28) Sub device error.		

BRM *BCS Report Mode:*

BRM			
This command reports the mode of the BCS.			
Parameter	***Validity***	***Default***	***Description***
None			
Response	***if***		
			Report the mode of the BCS. 0 → BCS in scan mode 1 → BCS beam on mode 2 → BCS in Set mode
Error	(5) Device not implemented, (11) Sub device not ready yet, (21) BCS communication error.		

BSM *BCS Set Mode:*

BSM Mode			
This command switches beam on mode. Then the beam of the BCS is permanently on, to adjust the position of the BCS. In this mode no further action with the BCS can be taken. The Sub device error (28) can be reported more detailed by the **REB** command. ATTENTION: In beam on mode the test button on the BCS should not be used.			
Parameter	***Validity***	***Default***	***Description***
Mode	0,1	0	0 → Set BCS to normal mode 1 → Set BCS to beam on mode
Error	(3) Invalid operand, (5) Device not implemented, (11) Sub device not ready yet, (21) BCS communication error, (28) Sub device error		

MMD *Multiple Download:*

MMD Hexfile line			
This command loads the Hexfile line down into the subdevices selected by the **SMD** command. If an error occurs in a subdevice, the subdevice will be excluded of fuhrter downloads. Report the download state of each subdevice with **RMD**. **Caution:** Before sending the first Hexfile line by this command, erase the old firmware on the subdevices to be downloaded.			
Parameter	***Validity***	***Default***	***Description***
Hexfile line	Intel Hex Format		Hex line to be downloaded into the selected subdevices
Error	(3) Invalid operand, (25) Download Error (general),		

4.7 Break Commands (ROMA Specific Functionality)

BMX *Break X-Drive Movement:*

BMX Selector			
This function stops the movement of the X-drive in the commands **PAX**, **PRX**, **MAX**, **MRX**, **PAA** and **MAA** immediately either with normal deceleration or with an emergency break. Additionally the axis can be powered down for position evaluation. After breaking a movement, a new positioning command can be send. The drive doesn't lose the position. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2	0	0 = normal deceleration, 1 = emergency break, 2 = power down drive.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

BMY *Break Y-Drive Movement:*

BMY Selector			
This function stops the movement of the Y-drive in the commands **PAY**, **PRY**, **MAY**, **MRY**, **PAA** and **MAA** immediately either with normal deceleration or with an emergency break. Additionally the axis can be powered down for position evaluation. After breaking a movement, a new positioning command can be send. The drive doesn't lose the position. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2	0	0 = normal deceleration, 1 = emergency break, 2 = power down drive.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

BMZ *Break Z-Drive Movement:*

BMZ Selector			
This function stops the movement of the Z-drive in the commands **PAZ**, **PRZ**, **MAZ**, **MRZ**, **PAA** and **MAA** immediately either with normal deceleration or with an emergency break. Additionally the axis can be powered down for position evaluation. After breaking a movement, a new positioning command can be send. The drive doesn't lose the position. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2	0	0 = normal deceleration, 1 = emergency break, 2 = power down drive.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

BMR *Break R-Drive Movement:*

BMR Selector			
This function stops the movement of the R-drive in the commands **PAR**, **PRR**, **MAR**, **MRR**, **PAA** and *MAA* immediately either with normal deceleration or with an emergency break. Additionally the axis can be powered down for position evaluation. After breaking a movement, a new positioning command can be send. The drive doesn't lose the position. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2	0	0 = normal deceleration, 1 = emergency break, 2 = power down drive.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

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BMG *Break G-Drive Movement:*

BMG Selector			
This function stops the movement of the G-drive in the commands **PAG**, **PRG**, **MAG**, **MRG**, **PAA** and *MAA* immediately either with normal deceleration or with an emergency break. Additionally the axis can be powered down for position evaluation. After breaking a movement, a new positioning command can be send. The drive doesn't lose the position. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Selector	0..2	0	0 = normal deceleration, 1 = emergency break, 2 = power down drive.
Error	(3) Invalid operand, (4) CAN ack problem, (6) CAN answer timeout.		

BMA *Break All Drive Movement:*

BMA			
This function stops all movements of all running axis immediately. Movements activated by the **PAA**, **MAA** and **AAA** command can be stopped. After breaking all axis movements, also the vector table entries are removed. The actual table position where the movements have been stopped can be reported by the **RAE** command. To access a running drive, this command must be sent via the alternative access channel (second access node).			
Parameter	***Validity***	***Default***	***Description***
Error	(4) CAN ack problem.		

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4.8 Broadcast Commands

ROMA Broadcast Commands Codes	
These broadcast commands can be used only in combination with the new TeCU hardware. All the commands listed below can be distributed by the two TeCU system commands **SCC** and **SBC**. The broadcast messages can be sent independently and simultaneously to the other normal commands. There is no answer returned by these commands. Broadcast commands have a common meaning to all devices implemented at TECAN. So the command list below must be exactly the same or a subset as like in other modules. CAUTION: This command should be used only by trained personal.	
Code	***Description***
0	Software reset.
4	Normal move break.
5	Emergency move break.
32	Turn keep-alive messages on.
33	Turn keep-alive messages off.

4.9 Error Codes ROMA

ROMA Errors Codes		
Code	***Description***	***Source***
1	Initialization failed.	Initialization commands i.e. PIX, PIA.
2	Invalid command.	Not available commands.
3	Invalid operand.	Parameter out of range.
4	CAN acknowledge problems.	Communication to local CAN bus.
5	Device not implemented.	Device connected to local CAN bus missing.
6	CAN answer timeout.	Communication to local CAN bus.
7	Device not initialized.	Move commands i.e. PAX, PAA.
8	Command overflow of TeCU	Communication pipeline PC – TeCU overflow.
9	Plate not fetched.	AGR command.
10	Drive no load.	Crash detected.
11	Sub device not ready yet	Sub device is booting up
13	No access to Flash-EPROM.	HW problem.
14	Hardware not defined	Hardware not defined
15	Command overflow of this device	Communication pipeline TeCU – DCU overflow.
17	Verification failed	Wrong or no barcode read
21	BCS communication error	Failure in BCS communication
25	Download Error	Error occurred while downloading a Hex line
28	Sub device error	Read detailed BCS errors by **REB** command
30	Invalid servo version	Old DC-Servo 1 or wrong DC-Servo 2 Version.

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4.10 Requirement Downwards Traceability

FRS	FCS (3letter Cmd)	Description (FCS)
7.1.1	*All commands*	*Communication Interface*
7.1.2	*All commands*	*Communication Command Types*
7.2.1	*SFV* *RED* *SED* *AWE* *ARE* *ACW*	*Set Serial Number* *Read FLASH-EPROM Data* *Set FLASH-EPROM Data* *Write System Parameters to FLASH-EEPROM* *Read System Parameters to FLASH-EPROM* *Erase All Setup Data in FLASH-EEPROM*
7.2.2	*RFV*	*Report Firmware Version*
7.2.3	*RHW* *SHW* *ROD* *SOD*	*Report ROMA Hardware Version* *Set ROMA Hardware Version* *Report Gripper Outrigger Distance* *Set Gripper Outrigger Distance*
7.2.4	*RDA*	*Report Device Allocation Array of System Devices*
7.2.5	*RGD* *SGD*	*Report Global Data* *Set Global Data*
7.3.1	*RDF* *RDX* *RDY* *RDZ* *RDR* *RDG* *SMC*	*Report Diagnostic Functions* *Report Diagnostic Functions for the X-Axis* *Report Diagnostic Functions for the Y-Axis* *Report Diagnostic Functions for the Z-Axis* *Report Diagnostic Functions for the R-Axis* *Report Diagnostic Functions for the G-Axis* *Set all Clearable ROMA Axis Move Counters to Zero*
7.4	*SOF* *RMD* *SMD* *MMD*	*Firmware Download* *Report Multiple Download* *Set Multiple Download* *Multiple Download*
7.5.1	*RFV*	*Report Firmware Version*
7.5.2	*PIA* *PIB* *PIF* *PIX* *PIY* *PIZ* *PIR* *PIG*	*Position Initialization X/Y/Z/R/G – Axis* *Position Pre Initialization* *Fake Initialization X/Y/Z/R/G – Axis* *Position Initialization X-Axis* *Position Initialization Y-Axis* *Position Initialization Z-Axis* *Position Initialization R-Axis* *Position Initialization G-Axis*
7.5.3	*REE* *RPX* *RPY* *RPZ* *RPR* *RPG* *RSM* *SSM* *SRX* *SRY* *SRZ* *SRR* *SRG* *SOX* *SOY* *SOZ* *SOR* *SOG*	*Report Extended Error Code* *Report Current Parameter for the X-Axis* *Report Current Parameter for the Y-Axis* *Report Current Parameter for the Z-Axis* *Report Current Parameter for the R-Axis* *Report Current Parameter for the G-Axis* *Report X-Axis Smooth Move Mode* *Set X-Axis Smooth Move Mode* *Set X-Axis Range* *Set Y-Axis Range* *Set Z-Axis Range* *Set R-Axis Range* *Set G-Axis Range* *Set Initialization Offset for X-Axis* *Set Initialization Offset for Y-Axis* *Set Initialization Offset for Z-Axis* *Set Initialization Offset for R-Axis* *Set Initialization Offset for G-Axis*

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	SSX	*Set Slow Speed for X-Axis*
	SSY	*Set Slow Speed for Y-Axis*
	SSZ	*Set Slow Speed for Z-Axis*
	SSR	*Set Slow Speed for R-Axis*
	SSG	*Set Slow Speed for G-Axis*
	SFX	*Set Fast Speed and Acceleration for X-Axis*
	SFY	*Set Fast Speed and Acceleration for Y-Axis*
	SFZ	*Set Fast Speed and Acceleration for Z-Axis*
	SFR	*Set Fast Speed and Acceleration for R-Axis*
	SFG	*Set Fast Speed and Acceleration for G-Axis*
	SAX	*Set Scale Adjust Factor for X-Axis*
	SAY	*Set Scale Adjust Factor for Y-Axis*
	SAZ	*Set Scale Adjust Factor for Z-Axis*
	SAR	*Set Scale Adjust Factor for R-Axis*
	SAG	*Set Scale Adjust Factor for G-Axis*
	SLX	*Set X-Axis Scaling Factor and Motion Direction*
	SLY	*Set Y-Axis Scaling Factor and Motion Direction*
	SLZ	*Set Z-Axis Scaling Factor and Motion Direction*
	SLR	*Set R-Axis Scaling Factor and Motion Direction*
	SLG	*Set G-Axis Scaling Factor and Motion Direction*
	PAX	*Position Absolute X-Axis*
	PAY	*Position Absolute Y-Axis*
	PAZ	*Position Absolute Z-Axis*
	PAR	*Position Absolute R-Axis*
	PAG	*Position Absolute G-Axis*
	PRX	*Position Relative X-Axis*
	PRY	*Position Relative Y-Axis*
	PRZ	*Position Relative Z-Axis*
	PRR	*Position Relative R-Axis*
	PRG	*Position Relative G-Axis*
	MAX	*Position Absolute with Slow Speed X-Axis*
	MAY	*Position Absolute with Slow Speed Y-Axis*
	MAZ	*Position Absolute with Slow Speed Z-Axis*
	MAR	*Position Absolute with Slow Speed R-Axis*
	MAG	*Position Absolute with Slow Speed G-Axis*
	MRX	*Position Relative with Slow Speed X-Axis*
	MRY	*Position Relative with Slow Speed Y-Axis*
	MRZ	*Position Relative with Slow Speed Z-Axis*
	MRR	*Position Relative with Slow Speed R-Axis*
	MRG	*Position Relative with Slow Speed G-Axis*
	PAA	*Position Absolute for all Axis*
	PRA	*Position Relative for all Axis*
	MAA	*Position Absolute with Slow Speed for all Axis*
	MRA	*Position Relative with Slow Speed for all Axis*
	ARX	*Auto Range X-Axis*
	ARY	*Auto Range Y-Axis*
	ARZ	*Auto Range Z-Axis*
	ARR	*Auto Range R-Axis*
	ARG	*Auto Range G-Axis*
	SPC	*Set Drive Parameter Classes*
	RPC	*Report Drive Parameter Classes*
	STW	*Set Target Window Classes*
	RTW	*Report Target*
	AAC	*Action Move to Vector with class attributes*
	RPT	*Report Permanent Target Window*
	STP	*Set Permanent Target Window*
7.5.4	*RGG*	*Report Gripper Parameters*
	SGG	*Set Gripper Parameters*
	AGR	*Grip a Plate*
7.5.5	*RAA*	*Report Vector Coordinate Position*

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	RAE *SAA* *SRA* *AAA*	*Report Actual Vector Number of AAA Movement* *Set Vector Coordinate Position into Table* *Remove Vector Coordinate Position into Table* *Action Move to Vector Coordinate Position*
7.5.6	*RBK* *SBK*	*Report Z-Axis Break Status* *Set Z-Axis Break On or Off*
7.5.7	*RIB* *REB* *BRM* *SIB* *BSM* *TRB* *ABR* *ARC*	*Report baud rate of BCS* *Report error status of BCS* *Report mode of BCS* *Set baud rate of BCS* *Set mode of BCS* *Transfer to barcode scanner* *Read barcode* *Read reference barcode*
9.1	*All commands*	*Communication*
9.2	*SOF*	*Firmware Download* *Watchdog solved in electronics HW*
9.3	*SCC* *SBC* *BMA* *BMX* *BMY* *BMZ* *BMR* *BMG*	*Send Broadcast Message to specific Slave Device* *Send Broadcast Message to all Slave Devices* *Break All Drive Movement* *Break X-Drive Movement* *Break Y-Drive Movement* *Break Z-Drive Movement* *Break R-Drive Movement* *Break G-Drive Movement*
9.5	*RSU* *RTX*	*Report Stack Usage* *Report Real Time Executive Parameters*
9.6	*BMX* *BMY* *BMZ* *BMR* *BMG* *SRS*	*Break X-Drive Movement* *Break Y-Drive Movement* *Break Z-Drive Movement* *Break R-Drive Movement* *Break G-Drive Movement* *Set Reset System*

4.11 Requirement Upwards Traceability

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AAC	7.5.3	Axis Movement & Parameter
ABR	7.5.7	BCS Handling
ACW	7.2.1	Device Configuration
AGR	7.5.4	Plate Gripper Function
ARC	7.5.7	BCS Handling
ARE	7.2.1	Device Configuration
ARG	7.5.3	Axis Movement & Parameter
ARR	7.5.3	Axis Movement & Parameter
ARX	7.5.3	Axis Movement & Parameter
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ARZ	7.5.3	Axis Movement & Parameter
AWE	7.2.1	Device Configuration
BMA	9.3	Security commands
BMG	9.3 9.6	Security commands Emergency Stop
BMR	9.3 9.6	Security commands Emergency Stop
BMX	9.3 9.6	Security commands Emergency Stop
BMY	9.3 9.6	Security commands Emergency Stop
BMZ	9.3 9.6	Security commands Emergency Stop
BRM	7.5.7	BCS Handling
BSM	7.5.7	BCS Handling
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MAR	7.5.3	Axis Movement & Parameter
MAX	7.5.3	Axis Movement & Parameter
MAY	7.5.3	Axis Movement & Parameter
MAZ	7.5.3	Axis Movement & Parameter
MMD	7.4	Firmware Download
MRA	7.5.3	Axis Movement & Parameter
MRG	7.5.3	Axis Movement & Parameter
MRR	7.5.3	Axis Movement & Parameter
MRX	7.5.3	Axis Movement & Parameter
MRY	7.5.3	Axis Movement & Parameter
MRZ	7.5.3	Axis Movement & Parameter
PAA	7.5.3	Axis Movement & Parameter
PAG	7.5.3	Axis Movement & Parameter
PAR	7.5.3	Axis Movement & Parameter
PAX	7.5.3	Axis Movement & Parameter
PAY	7.5.3	Axis Movement & Parameter
PAZ	7.5.3	Axis Movement & Parameter
PIA	7.5.2	Axis Initialization
PIB	7.5.2	Axis Initialization
PIF	7.5.2	Axis Initialization
PIG	7.5.2	Axis Initialization
PIR	7.5.2	Axis Initialization
PIX	7.5.2	Axis Initialization
PIY	7.5.2	Axis Initialization
PIZ	7.5.2	Axis Initialization

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PRA	7.5.3	Axis Movement & Parameter
PRG	7.5.3	Axis Movement & Parameter
PRR	7.5.3	Axis Movement & Parameter
PRX	7.5.3	Axis Movement & Parameter
PRY	7.5.3	Axis Movement & Parameter
PRZ	7.5.3	Axis Movement & Parameter
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RDZ	7.3.1	Parameter Storage in FLASH
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RED	7.2.1	Device Configuration
REE	7.5.3	Axis Movement & Parameter
RFV	*7.2.2* *7.5.1*	Firmware Identification Firmware Identification
RGD	7.2.5	Universal Data Array
RGG	7.5.4	Plate Gripper Function
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RMD	7.4	Firmware Download
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RPR	7.5.3	Axis Movement & Parameter
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